National Mobility Big Data & Spatial Intelligence Monograph • Official Guide (v1.3.1)

Taiwan Mobility Pulse

The Master Guide to National Multi-Modal Mobility & Spatial Intelligence

Synthesizing 448 million verified ticket tap transactions from MOTC TICP, millions of Freeway Bureau eTag gantry vehicle observations across 5 regulatory classes, TDX 74 scenic 4K/360° live CCTV streams, and CWA CODiS 30-year microclimate observations.
Deciphering 114 dynamic flow corridors, islandwide weather disruption tides, and scenic mountain-to-coast tourism pulses.

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Taiwan Mobility Pulse: The Master Guide to National Multi-Modal Mobility & Spatial Intelligence

國家級公共運輸票證與高速公路車流大數據全景指南

Language Switcher / 語言切換: [English] | 繁體中文版

> "The breathing of a city is inscribed in the card swipe records of the morning's first metro train; the pulse of a nation beats beneath the wheels of heavy freight trucks rushing along national freeways at midnight."
>
> **Full Master Monograph**: This guide has been compiled into a publication-grade interactive monograph (featuring 9 macro chapters, full-color data schemas, multi-dimensional comparison matrices, and direct teleportation portals):
>
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Preface: Why Do We Need an "Interactive Monograph"?

Historically, transportation and urban mobility big data projects have often fallen into two extremes:

1. **Extreme 1: Cold, Opaque Statistical Reports**
   Filled with impenetrable Origin-Destination (OD) tables, static matrices, and hundreds of pages of unnavigable PDFs. They lack spatial vitality—decision-makers cannot intuitively experience the temporal rhythms and territorial tides of human mobility.
2. **Extreme 2: Dazzling yet Superficial Command Dashboards**
   Floating neon particles and pulsing glow points offer a futuristic visual spectacle, but fail to answer fundamental analytical questions: **"Why were these datasets selected?", "What is the physical ground truth behind these numbers?", and "How do these data sources strictly exclude and complement one another?"**

Taiwan Mobility Pulse was created to bridge this chasm. By fusing **448 million real ticket transactions from MOTC TICP**, **millions of eTag freeway gantry vehicle observations from Freeway Bureau**, **real-time TDX parking and civic infrastructure APIs**, **30 years of continuous microclimate historical observations from CWA CODiS**, and **strategic cellular signaling frameworks**, we have synthesized a publication-grade interactive monograph.

This guide serves as your comprehensive chapter roadmap, leading you step-by-step through the most compelling mobility dynamics on the island of Taiwan. Every chapter includes **dedicated interactive map teleportation links** and **hands-on operational highlights**, enabling you to read, click, and verify in real time!

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Chapter Overviews

- **Chapter 1: Data Philosophy & Architectural Selection — Why "Rigid Skeleton + Flexible Muscle"?**
- An in-depth critique of relying solely on coarse cellular signaling. Explains why TICP ticket transactions and eTag freeway flows serve as the zero-drift physical ground truth. Analyzes identity truth labels, natural mutual exclusivity, and double-counting mitigation.
- **Chapter 2: National Data Foundation & Entity Schemas — TICP, eTag, TDX, and CODiS Specifications**
- Deconstructs 448 million ticketing records, 325 eTag gantries, TDX parking APIs, and CWA CODiS 30-year climatological datasets. Explains physical field definitions, temporal granularities, salted hash de-identification, and real JSON samples.
- **Chapter 3: Intercity Arteries & Urban Mobility Corridors — 114 Dynamic Flow Corridors**
- Maps High-Speed Rail (THSR), Taiwan Railway (TRA), 5 major metro systems, industrial tech freight corridors (Hsinchu Science Park, International Ports), and scenic highways. Analyzes weekday peak commuting vs. weekend migration tides.
- **Chapter 4: Hotspot Radiation & Off-Ramp Catchment Halos — eTag Flow Injection & Parking Pressure**
- Introduces the groundbreaking "Off-Ramp Discharge Catchment Halos". Traces vehicle dispersion into local urban grids across 16 strategic interchanges and proves strong correlation with TDX real-time parking lot occupancy.
- **Chapter 5: Four Micro-Personas & Weather Sensitivity — Who Moves? How Weather Reshapes Routes**
- Constructs four personas: Iron Commuters, Active Seniors, Gen-Z Students, and Weekend Tourists. Features the Weather Simulation Lab, revealing how storms induce a sharp plunge in cycling/bus trips while underground rail proves resilient.
- **Chapter 6: Islandwide Climate Pulse & Microclimate Livability — 30-Year Variations & Seasonal Tides**
- Interfaces with 30 years of CWA CODiS data. Unveils the Taipei basin +0.8°C heat island, precipitation condensation into intense cloudbursts, dehumidifier 24/7 overload rankings (Anbu 98, Linkou 77), natural typhoon shelters (Taichung 85), and a 12-month interactive climate player.
- **Chapter 7: Commercial Pulse & Station Hub Intelligence — TOD Catchments, EV Charging & Retail**
- Elevates transit nodes into TOD commercial lifestyle anchors. Empirically evaluates Taiwan's Top 30 transit hubs, linking customer spending propensity, EV 360kW supercharger gap radars, and grab-and-go meal peak hours.
- **Chapter 8: National Multi-Source Data Fusion Strategy — Door-to-Door Closed Loop & Policy Matrix**

- Formulates the "1 + 1 + 1 > 3" Door-to-Door full-day travel closed loop. Details the pioneering inverse calibration algorithm using hard ticketing/eTag anchors to rectify cellular signaling expansion bias, followed by a procurement decision matrix.
- **Chapter 9: Scenic Tourism, Live CCTVs & Holiday Traffic Dispersal — Real-Time Travel Intelligence**
- Fuses MOTC Tourism Administration's 74 4K/360° scenic live CCTVs, CWA CODiS climatology, Freeway eTag holiday surge multipliers (+320%), and 92 TaiwanTrip bus routes into a 4D decision framework for real-time crowd inspection, weather comfort, and congestion bypass.

Interactive Map Quick Index

Clicking the hyperlinked portals below opens the live web application in a new browser tab, automatically navigating to the corresponding interactive laboratory:

Chapter Theme	Core Analytical Concept	Live Web App Portal (Click to Open)	Key Highlights to Explore
Ch 1 Data Philosophy	Drift contrast, identity truth, mutual exclusivity	**Three-Source Strategic Matrix**	Inspect the Master Table 3-way matrix; compare pre-processing and audit costs
Ch 2 Data Dictionary	448M census baseline, field schemas, raw samples	**Data Foundation & Dictionary Matrix**	Review real JSON payloads for TICP ticketing, eTag 5 vehicle classes, and CODiS weather
Ch 3 Corridors	114 flow particle streams, freight vs. tourism	**Dynamic Flow Corridors Map (FlowMap)**	Toggle between "Workday" and "Weekend"; drag timeline to observe Hsinchu & Hsuehshan tides
Ch 4 Off-Ramp Halos	16 strategic interchanges, TDX parking pressure	**Spatiotemporal Heatmap & Catchment Halos**	Turn on " /:\ eTag Interchange Halos", click "Ailan / Sun Moon Lake" to see parking strain
Ch 5 Personas & Weather	4 micro-personas, Sunny vs. Torrential Rain	**Micro-Personas & Weather Simulation Lab**	Select "Active Seniors", switch to "Torrential Rain", observe elderly trip drop-off
Ch 6 Climate Pulse	30-year trends, extreme rankings, 12-month tides	**Climate Pulse & Livability Lab**	Activate "12-Month Seasonal Player", observe monthly temperature swings and mold index
Ch 7 Commercial Pulse	Top 30 transit hubs, EV supercharging gaps	**Commercial Pulse & Station Intelligence**	Select "Taipei Main Station" or "Banqiao" to view right HUD commercial diagnostics

Chapter Theme	Core Analytical Concept	Live Web App Portal (Click to Open)	Key Highlights to Explore
Ch 8 Global Fusion	Door-to-Door 5-stage loop, hard anchor calibration	**Multi-Source Fusion & Decision Guide**	Examine the Yonghe → Neihu travel reconstruction and TOD enterprise selection guide
Ch 9 Tourism & CCTV	74 4K/360° live scenic feeds, holiday surge	**Tourism Destinations & Live CCTV Dashboard**	Click "Live CCTV" to view mountain sea of clouds, CODiS best seasons & eTag surges

Reading & Exploration Recommendations

1. **Dual-Screen or Split-View Reading**: We recommend placing this guide on the left side of your screen and the interactive web application on the right. Verify empirical data on the map as you study the theoretical discourse.
2. **Follow the Practical Walkthroughs**: Each chapter contains step-by-step 【**Hands-on Operational Walkthroughs**】 and 【**In-Depth Data Takeaways**】 to help you quickly identify hidden patterns across millions of records.
3. **Engage with Investigative Inquiries**: Ponder questions such as "Why does heavy freight account for 34% of traffic at Daya Interchange on weekday mornings?" or "Does heavy rain truly leave metro ridership unaffected?" Detailed empirical validations await within.

Ready to begin? Let us turn to Chapter 1 and uncover the foundational philosophy behind Taiwan's transportation data!

Proceed to Next Chapter: Chapter 1: Data Philosophy & Architectural Selection →

Chapter 1: Data Philosophy & Architectural Selection — Why "Rigid Skeleton + Flexible Muscle"?

Language Switcher / 語言切換: [English] | 繁體中文版

> "In the realm of data science, vast volume does not equate to precision. If baseline coordinates are adrift and human identities are mere conjectures, even the most elaborate neural network merely magnifies spatial illusions."

1.1 The Eternal Dilemma of Big Data in Mobility: Noise & The Cost of Estimation

Over the past decade, intelligent transportation and human mobility analytics have frequently faltered due to a single blind spot: **unquestioned reliance on a single data source**.

Numerous initiatives attempted to solve all urban problems exclusively using **Cellular Signaling Data**: from metro ridership and street congestion to commercial district expenditure. However, when projects entered the stages of government audits, formal acceptance, or granular civil engineering, glaring discrepancies inevitably emerged:

- **Why were passengers standing on an underground platform 3 levels deep misclassified by cellular towers as driving cars on the elevated viaduct overhead?**
- **Why could statistical expansion figures generated from cellular signaling never be legally utilized to disburse billions of New Taiwan Dollars in government TPASS budget subsidies?**
- **Why, after an individual lingered continuously in a neighborhood for 8 hours, remained impossible to discern whether they were an office professional, a student, or an international tourist?**

The genuine hurdle in mobility data science has never been "whether the data volume is sufficiently large," but rather **the intrinsic physical and legal attributes of the data engineering pipeline**.

From its inception, this project established an uncompromising foundational philosophy:
"Anchor the baseline with zero-drift physical ground truth as the rigid skeleton, and drape continuous territorial coverage over it as the flexible muscle."

1.2 Why Choose TICP Public Transit Ticketing? (Centimeter-Level Rigid Ground Truth)

We adopted over **448 million records of real electronic ticketing transactions** from the Ministry of Transportation and Communications (MOTC) TICP platform as the core public transit foundation of this research. Its irreplaceable value manifests across three absolute dimensions:

1. **100% Physical Full Census (Census Baseline, Non-Sample Expansion)**:
   Every tap at a turnstile or bus farebox represents a physical legal and financial transaction. There is no synthetic statistical inflation factor (such as multiplying market share by 2.7×). The data represents the exact, unadulterated physical passenger count.

2. **Statutory Personas & Contractual Ground Truth**:
   Every ticketing record inherently carries a legally validated identity attribute:

3. `TicketType=4`: **TPASS Commuter Monthly Pass** (heavy regular commuters directly benefiting from national transit policies).
4. `TicketType=03/04`: **Senior & Concession Cards** (citizens aged 65+ or persons with disabilities receiving statutory municipal subsidies).
5. `TicketType=02`: **Student Cards** (middle, high school, and university students exhibiting distinctive morning drop-off and after-school cram school travel cycles).
6. `Single-Journey Token / Mobile QR`: **International tourists, leisure travelers, or occasional users**.
   Identities are objective legal facts, requiring zero probabilistic guesswork from machine learning classifiers.

7. **Centimeter-Level Platform & Vertical Height Precision (Z-Axis Truth)**:
   Fare turnstiles are physically bolted onto underground basement levels (B2, B3) or elevated platform structures. Regardless of severe urban canopy occlusion or multi-tier viaduct stacking, a turnstile tap establishes with 100% certainty that the passenger is physically inside the rapid transit system, eliminating elevation ambiguity.

1.3 Why Choose eTag Freeway Traffic Flows? (0m Gantry Freight & Passenger Separation)

For highway corridors and intercity travel, we integrated millions of freeway vehicle passage records from the Freeway Bureau's electronic toll collection (eTag) network:

1. **0m Gantry Precision with Zero Spatial Drift**:
   eTag microwave sensor gantries are mounted at fixed, precisely measured highway kilometer posts (e.g., National Freeway 1, Kilometer 23.4k). A vehicle detection event is tied to an invariant coordinate. Data processing requires only temporal timestamp normalization, eliminating the massive computational overhead of road-network map-matching.

2. **Classification Across 5 Statutory Vehicle Categories (The Lifeblood of Freight)**:
   The eTag infrastructure categorizes every passage based on official motor vehicle registration and axle counts into 5 statutory classes:

3. `Class 31`: Small Passenger Cars (private commuting, family outings).
4. `Class 32`: Small Freight Trucks (urban light logistics, plumbing/electrical trade vans).
5. `Class 41`: Large Passenger Buses (intercity coaches, tour buses).
6. `Class 42`: Heavy Freight Trucks (intercity arterial trunk freight).
7. `Class 51/52`: Articulated / Semi-Trailer Container Trucks (international seaports, petrochemicals, heavy steel).
   This illuminates Taiwan's real "industrial logistics skeleton"—a critical economic domain that transit ticketing cannot capture.

8. **Natural Mutual Exclusivity with Ticketing**:
   An individual cannot physically be driving a personal vehicle on a national freeway while simultaneously swiping a card inside an underground metro train at the exact same second. Therefore: $$\text{Total Intercity Corridor Flow } Q_{\text{Total}} = Q_{\text{eTag (Freeway Auto/Freight)}} + Q_{\text{Ticket (Rail/Transit)}}$$ **These two streams can be added directly together with zero double-counting**, establishing the cleanest gold standard for calculating **Modal Split**.

1.4 Why Integrate TDX Parking and Urban Functions?

Vehicles do not vanish into thin air upon descending an off-ramp.
When highway traffic pours into local arterial surface grids, the immediate shockwaves are absorbed by **local parking infrastructure capacity** and **commercial service networks**.

By interconnecting MOTC TDX real-time parking APIs and urban point-of-interest densities (EV chargers, convenience stores, hypermarkets) with interchange discharge flows, we seamlessly translate macro highway volumes into micro-scale urban carrying capacity indicators.

1.5 Limitations of Cellular Signaling: Spatial Drift, UID Scrambling, and Double-Counting

While cellular signaling offers comprehensive geographic coverage, it presents three severe vulnerabilities in transportation engineering:

1. **Spatial Drift & Ping-Pong Handover**:
   Cell tower coverage radii in urban areas span 200m to 500m. When a user is stationary inside a high-rise, their mobile phone frequently switches associations between neighboring base stations.

Algorithms easily misinterpret this jitter as high-speed movement. When elevated viaducts run directly above underground subway tunnels, cell towers cannot resolve the user's vertical elevation.

2. **Privacy Hash Rotations Fragmenting Trip Chains**:
   To comply with privacy regulations, telecommunications operators periodically scramble temporary pseudonymous identifiers (IMSI / temporary UIDs) on a daily or weekly basis, injecting stochastic spatial perturbation noise. **Once UIDs are randomized, tracking multi-week longitudinal trip chains becomes impossible**, erasing the distinction between loyal local commuters and one-off visitors.

3. **Severe Vehicle Cabin Double-Counting**:
   When passengers ride an underground metro train, cellular towers detect hundreds of mobile devices and apply a ~2.6× expansion factor. Simultaneously, metro faregates record every tap. If cellular estimates are added directly to ticketing records without calibration, **metro passenger volumes are dramatically inflated**. Deduplicating them without plain-text identity linkages requires probabilistic assumptions, introducing secondary estimation errors.

1.6 Strategic Paradigm: "Rigid Skeleton + Flexible Muscle" Three-Source Fusion

Synthesizing these empirical realities, this project established the national-level multi-source data fusion framework:

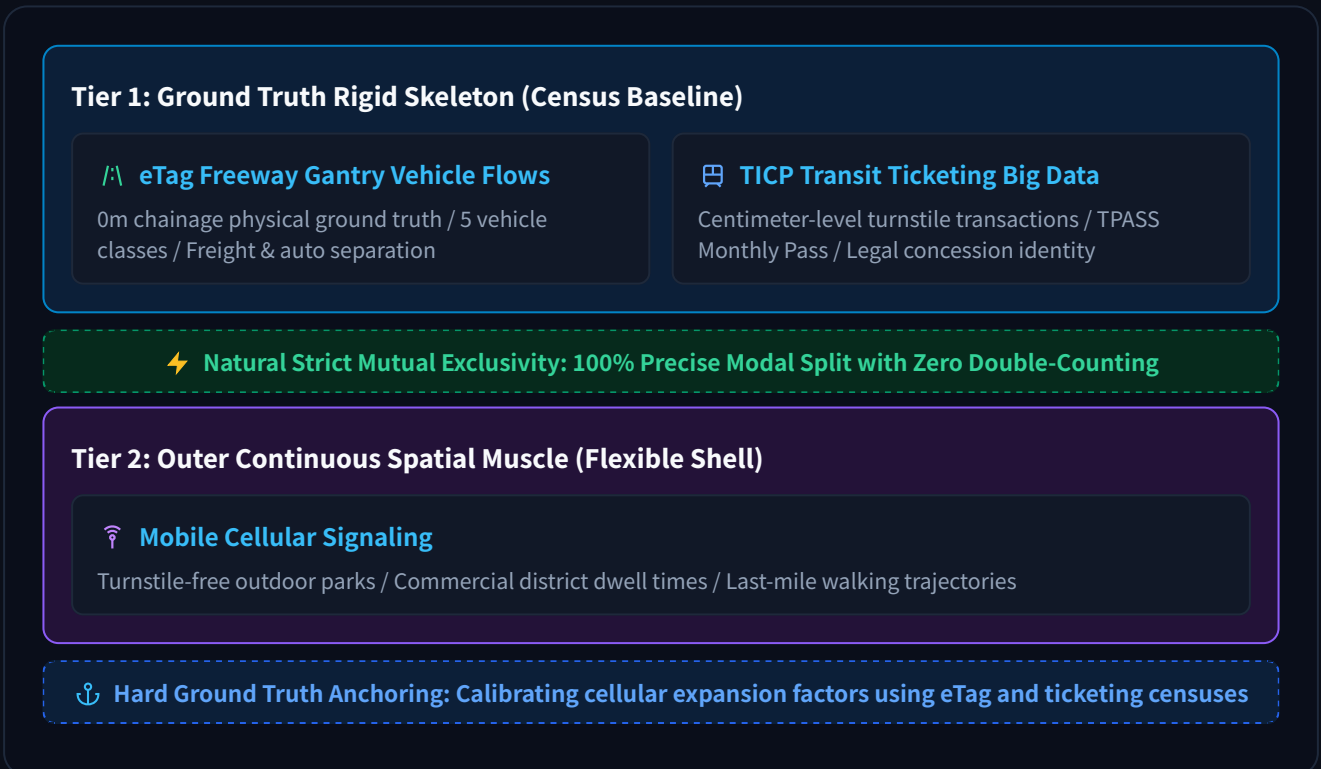

Figure 1.1: Live Application Interface — Three-Source Strategic Master Matrix illustrating the four core evaluation dimensions (Spatial Stability, Identity Contracts, Mutual Exclusivity, and Double-Counting Mitigation).

1.7 Hands-on Operational Walkthrough: Navigating the System

To experience the Three-Source Strategic Master Matrix firsthand, follow these steps:

OFFICIAL LIVE WEB APPLICATION PORTAL (CLICK TO OPEN)

Data Strategy Master Matrix →

Direct URL: https://tobytoy.github.io/taiwan-mobility-pulse/#telecom

Operational Steps:

1. Click the link above or click 【**Telecom vs. Ticketing**】 on the top navigation bar.
2. The system defaults to 【 **Act I: Strategic Master Matrix & Core Dimension Duel**】.

3. Review the four comparative duel cards (Spatial Physical Stability, Statutory Persona Contracts, Mutual Exclusivity, Activity Coverage & Redundancy).
4. Scroll down to inspect the **"Comprehensive Multi-Dimensional Cross-Comparison Table"**, contrasting 8 analytical indicators across eTag, TICP ticketing, and cellular signaling.

1.8 Core Data Takeaways & Analytical Inquiries

> **Thought Inquiry**:
> When government auditors evaluate whether the NT$ 10 billion annual public expenditure on the "Executive Yuan TPASS 1200 Commuter Pass" was justified: * Can statistical estimates from cellular signaling be submitted to the Directorate-General of Budget, Accounting and Statistics for expenditure write-offs?
> (No: cellular data lacks financial transaction ledgers, precise ticket class codes, and legal audit proofs.) * Why must TICP ticketing big data serve as the bedrock?
> (Because every single record represents an indelible, audited financial transaction.) * Why is eTag integration indispensable?
> (To substantiate whether private vehicle travel on parallel freeways truly decreased, and if so, exactly where that diverted traffic relocated.)

Having anchored our foundational data philosophy, we now proceed to dissect the lowest-level database architectures, field definitions, and raw schema payloads in the next chapter!

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Chapter 2: National Data Foundation & Entity Schemas — TICP Ticketing, eTag Gantries, TDX, and CODiS Specifications

Language Switcher / 語言切換: [English] | 繁體中文版

> "If one cannot discern which physical smart card, which microwave reflection from an overhead highway gantry, or which inductive loop in a parking garage a record represents, then even the most dazzling dynamic visualization remains nothing more than a drifting numerical mirage."

2.1 Macro Scope & Temporal Dimensions

"Taiwan Mobility Pulse" is not built upon localized survey sampling or probabilistic models from a single cellular carrier. Instead, it directly intersects and fuses three national-level ground truth data foundations:

National Mobility & Passenger Flow Data Infrastructure Overview

Data Category	Empirical Scale	Temporal Horizon & Sampling
MOTC TICP Public Transit Ticketing Base	**448 Million Verified Electronic Ticketing Taps** (843M Cross-Month Passenger Journeys)	**Continuous 181 to 255 Day Horizon** Second-level turnstile validations, hourly aggregates
Freeway Bureau eTag Electronic Toll Collection	**325 Gantry Posts & Interchanges Islandwide** 5 Statutory Motor Vehicle Regulatory Classes	**5-Minute Passage Aggregations** 24/7 continuous non-stop highway monitoring
MOTC TDX Parking & Amenities API	**76 Transit Hubs & 16 Strategic Interchange Radii** 3,000+ public and commercial parking facilities	**1 to 5 Minute Dynamic Refresh** Occupancy rates, turnover & step-tier tariffs
CWA CODiS Historical Climatology Base	**29 First-Order Meteorological Stations** Continuous 30-year hourly, daily & monthly series	**Hourly Continuous Observations** Sunny, moderate rain, torrential storm stress tests

Observation Temporal Baseline

1. **Macro Historical Time Horizon**:
   Encompasses continuous time-series observations across **H1 2024 through H1 2026**, covering **181 to 255 consecutive calendar days**. Includes normal academic school weeks (e.g., standard March terms), continuous workdays, Lunar New Year mass migrations, Tomb Sweeping festival, Dragon Boat festival, and Mid-Autumn festival periods.
2. **Dual Day-Type Classification**:
3. **Workdays**: Monday through Friday (strictly excluding national holidays and severe storm/typhoon shutdown days) to capture the 9-to-5 commuting peaks and industrial logistics pulses.
4. **Weekends**: Saturday and Sunday, isolating commercial leisure shopping, metropolitan outflows, and short-haul scenic excursions.
5. **Long Consecutive Holidays**: Breaks spanning 3+ days, isolating inter-regional family homecomings and mass tourism surges.
6. **Temporal Granularity**:
7. **Hourly (`00:00:00` ~ `23:00:00`)**: TICP public transit records are aggregated into hourly timestamps under privacy protection protocols.
8. **Transaction-Level (Seconds / Milliseconds)**: Tap-in and tap-out transactions, as well as 60-minute inter-modal transfer discount validations, are executed and resolved on smartcard chips with second-level timestamps.

2.2 TICP Transit Ticketing Data Dictionary

MOTC TICP consolidates 10 multimodal transit operators across Taiwan, structured into three primary schemas:

1. Urban Bus & Highway Coach Transaction Schema (`bus_to3a`)

The largest dataset capturing street-level capillary mobility (comprising 250M Taipei/New Taipei urban bus records and 26.6M intercity highway coach records):

Field Name	Type	Physical Definition & Business Logic
`LineID`	`VARCHAR(16)`	**Unique Route Identifier**: Authorized route code (e.g., `0100030700` denotes Route 307).
`SubLineID`	`VARCHAR(8)`	**Sub-Route Identifier**: Distinguishes mainline, shuttle, express, and jumping bus variants (`01` = mainline, `02` = shuttle).
`Direction`	`TINYINT`	**Travel Direction**: `0` = Outbound, `1` = Inbound.
`HolderType`	`CHAR(1)`	**Statutory Passenger Persona**: `A` General Adult Full Fare, `S` Student Concession, `E` Senior / Disability Concession, `K` Child Fare.
`CardType`	`TINYINT`	**Smartcard Issuer**: `1` EasyCard, `2` iPASS, `3` icash 2.0.
`TicketType`	`VARCHAR(16)`	**Fare Scheme Code**: `#NOR-1200` represents TPASS Northern 1200 Monthly Pass; `NORMAL` represents self-funded e-wallet.
`BoardingStopID`	`VARCHAR(16)`	**Boarding Stop ID**: Station code with exact geographical coordinates.
`BoardingStopSequence`	`SMALLINT`	**Boarding Stop Sequence**: Stop sequence number N along the trip path (1, 2, 3... N).
`BoardingTime`	`DATETIME`	**Boarding Timestamp**: De-identified hourly timestamp (e.g., `2026-03-11 08:00:00`).
`DeboardingStopID`	`VARCHAR(16)`	**Deboarding Stop ID**: Destination station stop code.
`DeboardingStopSequence`	`SMALLINT`	**Deboarding Sequence**: Destination sequence; the differential yields **total stops traveled**.
`DeboardingTime`	`DATETIME`	**Deboarding Timestamp**: Alighting transaction hourly timestamp.
`Fare`	`SMALLINT`	**Nominal Tariff (NT$)**: Statutory undiscounted fare (e.g., standard segment fare NT$ 15/30).
`Discount`	`SMALLINT`	**Statutory Subsidy Discount (NT$)**: Government subsidy or transfer concession (e.g., NT$ 8 transfer relief; NT$ 15 full TPASS waiver).
`Amount`	`SMALLINT`	**Net Amount Deducted (NT$)**: Actual cash deducted from card: `Amount = Fare - Discount` .
`TransferCode`	`VARCHAR(8)`	**Transfer Code**: `0` None, `102` Metro-to-Bus, `9902` Bus-to-Bus, `202` TRA-to-Bus.

Raw Ticketing JSON Samples


```
[
  {
    "LineID": "0100030700",
    "SubLineID": "01",
    "Direction": 0,
    "HolderType": "A",
    "CardType": 1,
    "TicketType": "#NOR-1200",
    "BoardingStopID": "TPE100341",
    "BoardingStopSequence": 14,
    "BoardingTime": "2026-03-11 08:00:00",
    "DeboardingStopID": "TPE100362",
    "DeboardingStopSequence": 23,
    "DeboardingTime": "2026-03-11 08:00:00",
    "Fare": 15,
    "Discount": 15,
    "Amount": 0,
    "TransferCode": "102"
  },
  {
    "LineID": "0100026500",
    "SubLineID": "01",
    "Direction": 1,
    "HolderType": "E",
    "CardType": 1,
    "TicketType": "NORMAL",
    "BoardingStopID": "NWT204112",
    "BoardingStopSequence": 6,
    "BoardingTime": "2026-03-11 09:00:00",
    "DeboardingStopID": "TPE101050",
    "DeboardingStopSequence": 18,
    "DeboardingTime": "2026-03-11 09:00:00",
    "Fare": 15,
    "Discount": 7,
    "Amount": 8,
    "TransferCode": "0"
  }
]
```


Sample Interpretation:
Sample 1: A TPASS 1200 Commuter (`HolderType='A'` , `TicketType='#NOR-1200'`) travels 9 stops on Route 307 during the 08:00 morning rush. Because the trip connects from a Taipei Metro exit (`TransferCode='102'`), the NT$ 15 fare is fully offset by the TPASS mechanism (`Discount=15`), resulting in `Amount=0` .
Sample 2: An **Elderly Citizen (`HolderType='E'`)** travels from Banqiao to Wanhua at 09:00 on Route

> 265, receiving a statutory concession discount of NT$ 7 and paying NT$ 8 (or redeemed via municipal monthly welfare points).

---

2. Rail Transit Hourly Origin-Destination Schema (`rail_od`)

Encompasses High-Speed Rail (THSR), Taiwan Railway (TRA), Taipei Metro (TRTC), New Taipei Metro (NTMC), and Kaohsiung Metro (KRTC), aggregating over 500 million journeys:

Field Name	Type	Physical Definition & Business Logic
`OriginStationID`	`VARCHAR(8)`	**Origin Station Code**: Turnstile entry code (e.g., TRTC `BL12` , TRA `1000` , THSR `02`).
`OriginStationName`	`VARCHAR(24)`	**Origin Station Name**: Canonical station name in Traditional Chinese.
`DestinationStationID`	`VARCHAR(8)`	**Destination Station Code**: Turnstile exit code (e.g., TRTC `BR15` , THSR `04`).
`DestinationStationName`	`VARCHAR(24)`	**Destination Station Name**: Canonical station destination name.
`TransactionDate`	`DATE`	**Travel Date**: Format `YYYY-MM-DD` .
`TransactionHour`	`TINYINT`	**Hourly Interval**: `0` ~ `23` (e.g., `8` denotes 08:00:00 to 08:59:59 alighting).
`PassengerCount`	`INTEGER`	**Passenger Volume**: Verified total swiped passengers across the specified OD pair in that hour.
`DayType`	`VARCHAR(12)`	**Calendar Type**: `Weekday` , `Weekend` , or `Holiday` .

Rail OD JSON Sample


```
[
  {
    "OriginStationID": "BL12",
    "OriginStationName": "Taipei Main Station",
    "DestinationStationID": "BR15",
    "DestinationStationName": "Xihu",
    "TransactionDate": "2026-03-12",
    "TransactionHour": 8,
    "PassengerCount": 3842,
    "DayType": "Weekday"
  },
  {
    "OriginStationID": "O2",
    "OriginStationName": "THSR Taipei",
    "DestinationStationID": "O4",
    "DestinationStationName": "THSR Hsinchu",
    "TransactionDate": "2026-03-12",
    "TransactionHour": 7,
    "PassengerCount": 1420,
    "DayType": "Weekday"
  }
]
```


3. Public Bikeshare Rental Schema (`bike_to2a`)

Captures first-and-last mile green mobility micro-circulation (40.4M Taipei/New Taipei YouBike records, 7.95M Taoyuan records):

Field Name	Type	Physical Definition & Business Logic
`RentStationID`	`VARCHAR(16)`	**Rental Station ID**: Unique YouBike dock station identifier.
`RentStationName`	`VARCHAR(48)`	**Rental Station Name**: Name and subway exit detail.
`RentTime`	`DATETIME`	**Rental Timestamp**: De-identified hourly timestamp (`YYYY-MM-DD HH:00:00`).
`ReturnStationID`	`VARCHAR(16)`	**Return Station ID**: Dock station identifier where bike was locked.
`ReturnStationName`	`VARCHAR(48)`	**Return Station Name**: Canonical return station name.
`ReturnTime`	`DATETIME`	**Return Timestamp**: Alighting hourly timestamp.
`DurationSeconds`	`INTEGER`	**Trip Duration (Seconds)**: Exact duration from unlock to dock latching.

2.3 Freeway Bureau eTag Toll Collection Schema

The Freeway Bureau's eTag network is Taiwan's sole dataset combining **statutory vehicle classification** with **fixed physical gantry chainage**:

Field Name	Type	Physical Definition & Business Logic
`GantryID`	`VARCHAR(16)`	**Gantry ID**: Official identifier (e.g., `01F-0232S` = Freeway 1 Southbound 23.2K Yuanshan).
`HighwayName`	`VARCHAR(16)`	**Freeway Name**: National Freeway 1, Freeway 3, Freeway 5, Freeway 6.
`MilestoneKM`	`DECIMAL(5,1)`	**Kilometer Post**: Meter-accurate physical chainage (e.g., `23.2`).
`InterchangeName`	`VARCHAR(24)`	**Associated Interchange**: Off-ramp exit name (e.g., Yuanshan, Hsinchu Science Park, Ailan).
`DischargeTarget`	`VARCHAR(64)`	**Discharge Catchment Area**: Urban district, technology park, or scenic corridor receiving the traffic flow.
`VehicleType`	`SMALLINT`	**Statutory Vehicle Class**: • `31`: Small Passenger Cars (Sedan/SUV) • `32`: Small Freight Trucks (<3.5T) • `41`: Large Passenger Buses (Coaches/Tour Buses) • `42`: Heavy Freight Trucks • `51/52`: Articulated / Semi-Trailer Container Trucks
`WorkdayDischargeVol`	`INTEGER`	**Weekday Daily Discharge (Vehicles)**: Normal workday 24-hour total.
`HolidayDischargeVol`	`INTEGER`	**Holiday Daily Discharge (Vehicles)**: Weekend/Holiday 24-hour total.
`FreightRatio`	`DECIMAL(4,1)`	**Heavy Freight Ratio (%)**: Proportion of classes 32+42+51/52 (identifies industrial supply chains).

Raw eTag Interchange JSON Sample


```
[
  {
    "GantryID": "01F-0955S",
    "HighwayName": "National Freeway 1",
    "InterchangeName": "Hsinchu Science Park Interchange (95K)",
    "DischargeTarget": "Hsinchu Science Park (Park 2nd Rd / Jieshou Rd)",
    "WorkdayDischargeVol": 78200,
    "HolidayDischargeVol": 22100,
    "VehicleMix": {
      "type31_car": 62,
      "type32_small_truck": 16,
      "type41_tour_bus": 4,
      "type42_big_truck": 12,
      "type51_52_trailer": 6
    },
    "PeakHour": "08:00 - 09:00",
    "PeakHourlyDischarge": 9840
  },
  {
    "GantryID": "06F-0290E",
    "HighwayName": "National Freeway 6",
    "InterchangeName": "Ailan Interchange (29K)",
    "DischargeTarget": "Sun Moon Lake National Scenic Area / Puli Downtown (Provincial Highway 14/21)",
    "WorkdayDischargeVol": 11200,
    "HolidayDischargeVol": 41800,
    "VehicleMix": {
      "type31_car": 82,
      "type32_small_truck": 6,
      "type41_tour_bus": 10,
      "type42_big_truck": 2,
      "type51_52_trailer": 0
    },
    "PeakHour": "10:00 - 12:00",
    "PeakHourlyDischarge": 5420
  }
]
```

2.4 TDX Real-Time Parking & Civic Infrastructure Dictionary

Field Name	Type	Physical Definition & Business Logic
`ParkingLotID`	`VARCHAR(32)`	**Unique Parking Facility ID**: Registered facility code in MOTC TDX.
`ParkingLotName`	`VARCHAR(64)`	**Facility Name**: Legal registered name.
`TotalSpaces`	`INTEGER`	**Total Capacity**: Authorized passenger car parking slots.
`AvailableSpaces`	`INTEGER`	**Real-Time Available Spaces**: Dynamic count transmitted via sensor loops / ALPR cameras.
`OccupancyRate`	`DECIMAL(4,2)`	**Dynamic Occupancy Ratio (0.00 ~ 1.00)**: $1 - (\text{AvailableSpaces} / \text{TotalSpaces})$; >0.85 indicates severe congestion.
`HourlyRate`	`SMALLINT`	**Hourly Tariff (NT$/hr)**: Standard hourly charging rate.
`DistanceToDischargeKM`	`DECIMAL(4,2)`	**Distance to Off-Ramp (km)**: Spatial buffer distance from highway discharge point.

2.5 CWA CODiS Climate Big Data Specifications

Climate represents the primary exogenous disruptor reshaping surface human flows and travel impedance. Through the Central Weather Administration's (CWA) official **Climate Observation Data Inquire Service (CODiS)** OpenAPI, this project connects with 30 years (1998–2024) of continuous hourly, daily, and monthly microclimate historical observations.

1. Station Hierarchy

- **Manned First-Order Stations (29 Stations)**: Historical stations with 30+ to 100+ years of rigorous manual calibration (e.g., Taipei 466920, Keelung 466940, Anbu 466910, Taichung 467490, Tainan 467410, Hengchun 467590, Alishan 467530, Yushan 467550, Penghu 467350), possessing full-spectrum meteorological instrumentation and authoritative public trust.
- **Automatic Weather Stations (C0 Series, 1,236 Stations)**: Distributed across rural townships, coastal capes, and mountain corridors, providing high spatial resolution temperature, humidity, and precipitation monitoring.

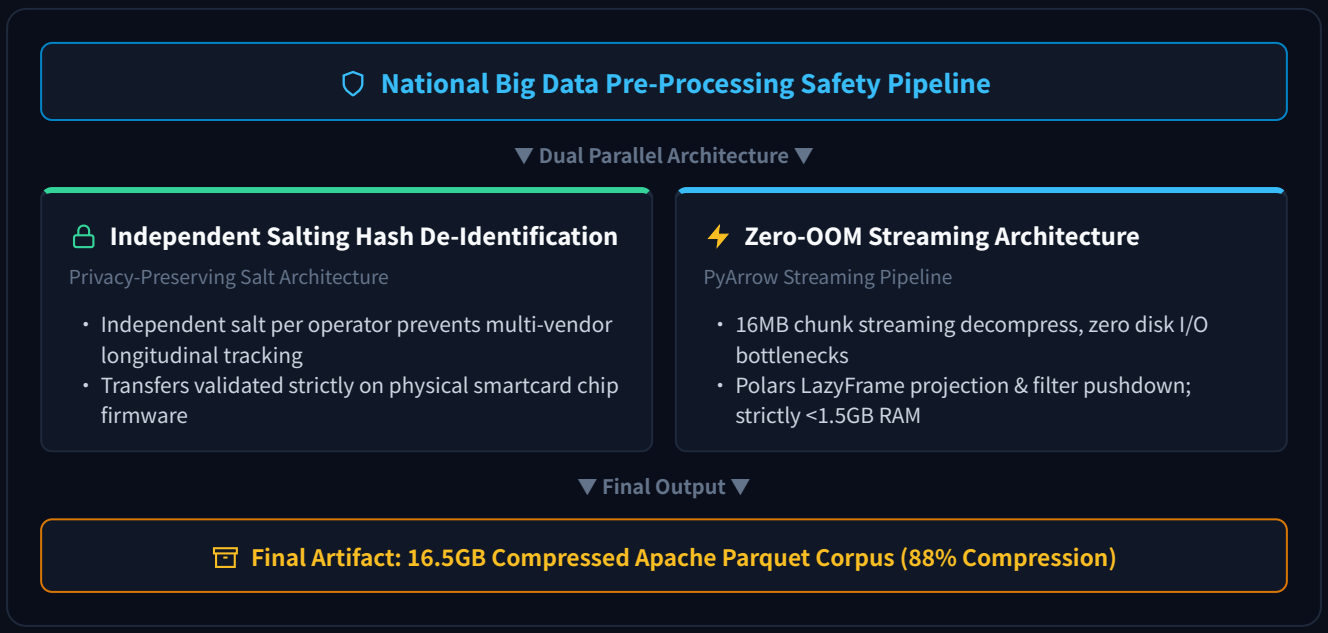
2. CODiS Field Dictionary

Field Name	Type	Definition & Calculation Logic
`StationID`	`VARCHAR(10)`	**WMO International Station ID**: e.g., `466920` (Taipei), `467490` (Taichung).
`StationName`	`VARCHAR(32)`	**Canonical Station Name**: Registered Chinese/English station name.
`County`	`VARCHAR(16)`	**Administrative District**: Municipality or County of installation.
`AltitudeM`	`DECIMAL(6,1)`	**Altitude (m)**: From 6.3m (Taipei) to 3,844.8m (Yushan), determining vertical temperature lapse rates.
`Archetype`	`VARCHAR(32)`	**Microclimate Persona**: 22 archetypes including "Heat Island Basin", "Moody Rainy Port", "Typhoon Shelter", "Global Wind Pole".
`TempDelta30Y`	`DECIMAL(4,2)`	**30-Year Temperature Net Shift (°C)**: Recent decade mean (2015–2024) minus baseline mean (1998–2007); national average +0.72°C.
`Hot35DaysDelta`	`DECIMAL(5,1)`	**Net Change in Extreme Heat Days (Days/Year)**: 30-year increase in days with maximum temperature $\ge 35^\circ\text{C}$ (+21.9 days in Taipei).
`RainDaysDelta`	`DECIMAL(5,1)`	**Net Change in Rainy Days (Days/Year)**: Trend reflecting fewer rainy days accompanied by more violent cloudburst intensities.
`MoldIndex`	`TINYINT`	**Indoor Mold Growth Index (0–100)**: Normalized score of annual days with relative humidity $>80\%$ and temperature between $20\sim 30^\circ\text{C}$.
`TyphoonDefense`	`TINYINT`	**Natural Typhoon Defense Score (0–100)**: Composite sheltering score based on Central Mountain Range lee shielding, historical peak gusts, and pressure drops.
`UmbrellaProbPct`	`DECIMAL(4,1)`	**Commute Umbrella Probability (%)**: Probability of precipitation during peak hours (07:00–09:00, 17:00–19:00).
`AnnualSunshineH`	`DECIMAL(6,1)`	**Annual Sunshine Hours (Hours/Year)**: Direct solar irradiance duration; evaluates photovoltaic and sunshine assets.
`AnnualRainMM`	`DECIMAL(6,1)`	**Annual Precipitation (mm/Year)**: Historical mean annual rainfall (3,828mm in Keelung vs. ~980mm in Penghu).
`MonthlyClimatology`	`JSON`	**12-Month Climatological Array**: Monthly temperature, rainfall, rain days, sunshine, humidity, and weather glyphs.

Raw CODiS Station Payload Sample


```
{
  "id": "466920",
  "name": "Taipei",
  "county": "Taipei City",
  "lat": 25.0377,
  "lng": 121.5149,
  "alt_m": 6.3,
  "archetype": "Heat Island Basin",
  "badge": "Taipei Basin • Champion of Nocturnal Warming",
  "metrics_30y": {
    "baseline_temp": 23.40,
    "recent_temp": 23.94,
    "temp_delta": 0.54,
    "hot35_days_delta": 21.9,
    "rain_days_delta": -10.4
  },
  "livability": {
    "avg_temp": 23.6,
    "annual_rain_mm": 2405.0,
    "annual_rain_days": 158.5,
    "annual_sunshine_h": 1305.2,
    "avg_humidity_pct": 75.8,
    "umbrella_prob_pct": 43.4,
    "mold_index": 58,
    "radar": {
      "rain_defense": 45,
      "thermal_comfort": 62,
      "typhoon_defense": 65,
      "sunshine": 48,
      "anti_mold": 42,
      "night_cool": 38
    }
  },
  "monthly_climatology": [
    { "month": 1, "temp": 16.7, "rain_mm": 88.0, "icon": "  ", "label": "Overcast" },
    { "month": 7, "temp": 30.1, "rain_mm": 245.2, "icon": " ", "label": "Clear & Sunny" }
  ]
}
```

2.6 Data Pipeline Nuances & Security Protections



1. **Independent Salting Hash Mechanism**:
   To safeguard personal privacy, TICP mandates that operators (Bus, Metro, TRA, YouBike) utilize **independent, distinct salting algorithms** when hashing smartcard numbers. As a result, analysts cannot link an individual's cross-system identity through simple card-number string joins.
   However, inter-modal transfer relationships are validated physically inside the onboard chip reader firmware, outputting verified statutory audit fields (`TransferCode` , `Discount`).
2. **Zero-OOM Streaming Architecture**:
   The raw uncompressed volume of 448 million records exceeds 30 GB, requiring over 120 GB of RAM if loaded with standard Pandas—prompting catastrophic Out-Of-Memory (OOM) crashes.
   Our pipeline (`pipeline/convert_to_parquet.py`) employs **PyArrow 16MB zero-disk stream decompression** combined with **Zstandard (Level 5) columnar compression**, shrinking disk footprint by 88% down to 16.5 GB. Using **Polars LazyFrames**, thread and memory consumption are constrained to a safe 1.5GB RAM ceiling!

Figure 2.1: Live Application Interface — Multi-Source Fusion Blueprint illustrating Tier 1 (Rigid Baseline), Tier 2 (Continuous Cellular Muscle), and Tier 3 (Inverse Anchoring Calibration).

2.7 Interactive Portal

To explore the multi-dimensional strategic comparison matrix synthesized from these 448 million records and 325 eTag gantries:

OFFICIAL LIVE WEB APPLICATION PORTAL (CLICK TO OPEN)

Strategic Master Matrix →

Direct URL: https://tobytoy.github.io/taiwan-mobility-pulse/#telecom

Having dissected the physical foundation and schemas, we now inject these microscopic flow particles into Taiwan's 114 representative mobility corridors!

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Chapter 3: Intercity Arteries & Urban Mobility Corridors — 114 Dynamic Flow Corridors

Language Switcher / 語言切換: [English] | 繁體中文版

> "If Taiwan were a living, breathing organism, then its 114 dynamic mobility corridors would be its arteries and veins; these coursing particles are not mere computer code, but the daily life journeys of twenty-three million souls."

3.1 Corridor Network Principles: From Isolated Stations to Connected Dimensions

Traditional transportation statistics often dwell upon isolated leaderboards of single stations (e.g., "Taipei Main Station ranks #1; Banqiao ranks #2"). However, point-based metrics cannot answer the core questions of urban mobility: **Where do people originate? Where are they bound? Which critical bottlenecks do they traverse?**

This system transcends point-based limitations. By intersecting Origin-Destination (OD) pairs from TICP electronic ticketing with contiguous passage paths across eTag highway gantries, we synthesized **114 of Taiwan's most representative dynamic mobility corridors**.

These corridors span five major multimodal transit categories: 1. **High-Speed Rail (THSR)**: The 350-kilometer western corridor backbone enabling the one-day living sphere. 2. **Taiwan Railway (TRA)**: The round-island intercity trunk line and regional commuter lifeblood. 3. **Metropolitan Rail Transit (Metro)**: High-density urban circulation across Taipei, New Taipei, Taoyuan, Taichung, and Kaohsiung. 4. **Intercity Highway Coaches**: Direct point-to-point commuter buses (e.g., Keelung ↔ Taipei; Nankan ↔ Taipei). 5. **Freeway Freight & Tourism Expressways**: High-capacity industrial logistics and scenic leisure routes monitored via eTag.

3.2 Industrial Logistics Arteries: Sleepless Heavy Corridors

Taiwan is an international epicenter for semiconductor fabrication and precision hardware. Beyond daytime commuter crowds, a massive undercurrent pulses across the island: **Industrial Freight**

Corridors. By analyzing passage records for eTag vehicle classes 32 (small trucks), 42 (heavy trucks), and 51/52 (tractor-trailers), we reconstructed Taiwan's three vital tech logistics corridors:

1. **Freeway 1 Silicon Corridor (Hsinchu Science Park ↔ Port of Taipei / Taoyuan Airport)**:
2. **Trajectory**: Hsinchu Science Park (HSP) → Freeway 1 / Provincial Highway 66 → Port of Taipei & Farglory Free Trade Zone.
3. **Characteristics**: 24-hour expedited transit for high-value semiconductor wafers and precision components. Between 00:00 and 04:00 at night, heavy box vans maintain remarkably high traffic densities.
4. **Central Port-Science Linked Corridor (Port of Taichung ↔ Central Taiwan Science Park)**:
5. **Trajectory**: Port of Taichung Container Terminals → Provincial Highway 74 / Freeway 4 → Central Taiwan Science Park (CTSP) Taichung Base.
6. **Characteristics**: Specialized transport for offshore wind turbine components, wafer foundry chemical tankers, and precision machinery export.
7. **Southern Semiconductor S-Corridor (Port of Kaohsiung ↔ Southern Taiwan Science Park)**:
8. **Trajectory**: Port of Kaohsiung Qianzhen Dedicated Freight Way → Freeway 1 / Provincial Highway 86 → Southern Taiwan Science Park (STSP) Xinshi.
9. **Characteristics**: Upstream chemical reagents and 3nm advanced fabrication supply chain deliveries.

Figure 3.1: Live Application Interface — Flow corridors filtered by "🚚 Industrial Logistics", highlighting the Freeway 1 Silicon Corridor and Port-Science logistics links operating around the clock.

3.3 Holiday Tourism Expressways: The Great Exodus to Mountains and Seas

In stark contrast to workday commuting, weekends and national holidays trigger a dramatic **"Polarity Inversion"** across Taiwan. Crowds rapidly evacuate metropolitan cores, surging into four national scenic corridors:

1. **Freeway 5 Hsuehshan Tunnel Yilan Express (Nangang ↔ Jiaoxi / Luodong)**:
2. While serving regional commuters on weekdays, **southbound traffic surges by +280% on Saturday mornings (07:00–11:00)**, while Sunday afternoons (14:00–20:00) produce northbound vehicle queues extending dozens of kilometers.
3. **Freeway 6 Ailan Sun Moon Lake Corridor (Wufeng → Puli / Ailan)**:
4. The primary artery funneling central metropolitan crowds toward Cingjing Farm, Sun Moon Lake, and Hehuanshan; small passenger cars (Class 31) account for over 88% of holiday traffic.
5. **South-Link Tourism Corridor (Pingtung Nanzhou → Kenting Hengchun)**:
6. The southern holiday trunk line; southbound flows swell from Friday evening, returning toward Pingtung and the Freeway 3 Nanzhou Interchange late Sunday.
7. **East Rift Valley Leisure Express (Taitung → Jiaoxi / Hualien)**:
8. Operated jointly by TRA EMU3000 New Ziqiang expresses and private driving along the Suhua Highway Improvement project.

Figure 3.2: Live Application Interface — Switched to "Holiday (Weekend)" and "Tourism" personas, showing a 320% explosion in particle flows across Freeway 5, Freeway 6, and the East Coast.

3.4 24-Hour Spatiotemporal Tidal Rhythm

Dragging the 24-hour timeline in the live system reveals Taiwan's daily mobility heartbeat:

```
00:00 ——— 05:00 | Nocturnal Rest Period: Transit halted; highway heavy freight (42/51) dominates
07:00 ——— 09:00 | Morning Rush Surge: Radial influx toward Taipei, Banqiao, HSP, Taichung, Kaohsiung
11:00 ——— 14:00 | Midday Business & Social Mobility: Short-haul trips within district centers, lunch
17:00 ——— 19:30 | Evening Commuter Exodus: Outward rush from urban centers; particle speeds peak
21:00 ——— 23:30 | Evening Homecomings & Night Markets: Dispersal from Xinyi, Fengjia, and Liuhe markets
```

3.5 Hands-on Operational Walkthrough: Experiencing the Dynamic FlowMap

OFFICIAL LIVE WEB APPLICATION PORTAL (CLICK TO OPEN)

Dynamic FlowMap →

Direct URL: https://tobytoy.github.io/taiwan-mobility-pulse/#flows

Operational Steps:

1. Click the link above or click 【**GIS Map**】 → 【**Dynamic FlowMap**】 on the top navigation bar.
2. **Observe the Timeline Tides**:
3. Locate the timeline control bar at the bottom and click **"Play"** to watch the hour advance from 00:00 to 23:00.
4. Notice the surging density of neon particles during the 08:00 morning rush and 18:00 evening peak.
5. **Toggle "Workday" vs. "Weekend"**:
6. Switch the day type on the top or left control panel:
7. Under **Workdays**, observe rapid blue particle influxes along Keelung → Taipei, the Hsinchu tech basin, and metropolitan subway webs.
8. Under **Weekends**, observe brilliant golden and orange leisure particles exploding along Freeway 5, Freeway 6, and the East Coast.
9. **Focus with the Corridor Quick Radar**:
10. Select **"HSP Tech Corridor"** or **"Freeway 5 Yilan Express"** in the corridor navigation list;
11. The camera will smoothly FlyTo and zoom into the corridor, highlighting its Origin-Destination coordinates and particle speeds!

3.6 Core Data Takeaways & Analytical Inquiries

> **Analytical Insights**: 1. **Morning Peak 08:00 on the "HSP Tech Corridor"**:
> Observe the vehicle particle stream converging onto the Hsinchu Interchange from Freeway 1, contrasted with simultaneous TRA Liujia branch train arrivals, demonstrating the multimodal balance between private driving and rail transfers. 2. **Evening Peak 18:00 on "Keelung → Taipei Highway Coaches"**:
> Notice the reverse flood of commuter coach particles departing Taipei Bus Station and City Hall Bus Station toward Keelung, illustrating the massive intercity capacity unlocked by the TPASS policy.

Having observed macro corridor flows, we must now investigate: when vehicles depart freeway off-ramps and enter local streets, how does local urban space absorb the shockwaves?

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Chapter 4: Hotspot Radiation & Off-Ramp Catchment Halos — eTag Flow Injection & Parking Pressure Interlocking

Language Switcher / 語言切換: [English] | 繁體中文版

> "A freeway is a high-pressure water pipe, and its off-ramps are the faucets. When a faucet is opened wide, the resulting shockwaves ripple down the ramp, inundating the surrounding urban ecosystem."

4.1 Beyond the Off-Ramp: The "Last Five Kilometers" Forgotten by Traditional Analytics

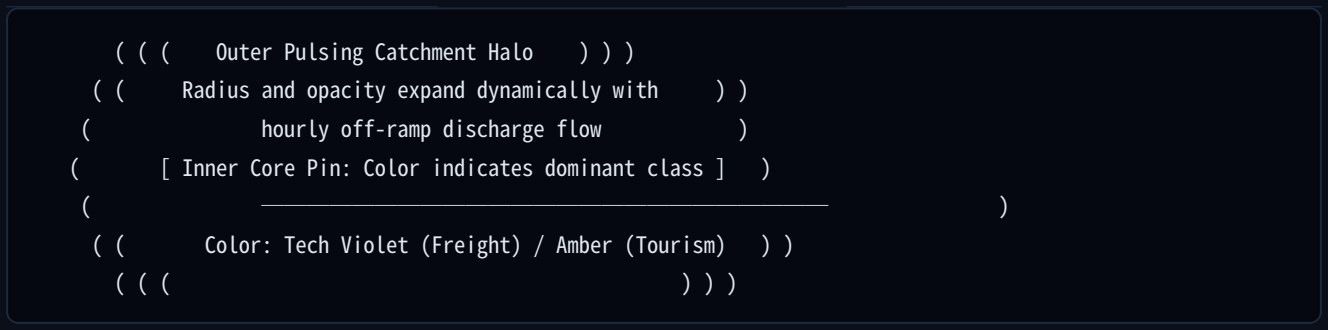
Historically, freeway traffic planning suffered from a severe analytical blind spot:

- The **Freeway Bureau** focused exclusively on **mainline congestion** (average vehicle speeds, bottleneck mileage markers).
- Municipal **Transportation Departments** focused solely on **intersection wait times and local parking stall counts**.
- **There was no dynamic linkage between the two**: When traffic pours down an off-ramp at 3,000 vehicles per hour, how much instant pressure is placed upon the carrying capacity, curbside spaces, and parking garages across the surrounding 3-to-5 kilometer urban radius?

To resolve this divide, our system introduced an innovative breakthrough: **【Scheme 1: Off-Ramp Discharge Catchment Halos】**.

4.2 Core Algorithm: Dual-Ring Dynamic Catchment Halos

Leveraging a hybrid Leaflet Canvas/SVG rendering engine, we engineered high-tech "breathing halos" for **16 strategic freeway interchanges** across Taiwan:



1. Dual-Ring Architectural Design

- **Outer Ripple Halo**:
  Driven by CSS `@keyframes etagHaloRipple` to execute continuous radial expansions. Its radius is computed dynamically from real-time off-ramp discharge volume (veh/h) scaled via a square-root transform (Scale Factor: 8~42px), directly visualizing the spatial dispersion potential onto surrounding street networks.
- **Inner Core Pin**:
  Anchors the exact geographical coordinates of the interchange. Border and core fills represent the dominant trip purpose (Green = suburban commuting, Violet = heavy industrial freight, Amber = scenic tourism).

2. Spatiotemporal Dynamic Scaling Formula

Halo radii combine volume metrics with calendar day-type attributes: $$R(t) = R_{\text{base}} \times \sqrt{\frac{V(t)}{V_{\text{max}}}} \times \gamma(\text{DayType})$$

- **Weekday Morning Rush**: Halos around **Hsinchu Science Park**, **Wugu**, and **Linkou** swell dramatically to their daily maxima.
- **Weekend Afternoons**: Halos around **Jiaoxi**, **Ailan (Sun Moon Lake)**, and **Daxi** expand by up to 2.5×, visually illustrating tourism bottlenecks queuing at highway exits.

4.3 Comprehensive Functional Profiles of 16 Strategic Interchanges

Based on vehicle classifications, hinterland attributes, and TDX historical parking saturation, Taiwan's 16 strategic interchanges exhibit distinct functional specializations:

Interchange Name	Highway	Functional Classification	Dominant Vehicle Mix	TDX Parking Pressure Characteristics
09A Xizhi	Freeway 1	Northern Metro Transit Knot	Cars 68%, Small Trucks 22%	Morning peak garages at 88% capacity
23 Yuanshan	Freeway 1	Downtown Taipei Gateway	Cars 78%, Coaches 14%	Jianguo Viaduct backups, full-day high load
33 Wugu	Freeway 1	New Taipei Industrial / Expwy Confluence	Heavy Trucks 24%, Semi-Trailers 12%	High loading dock and freight yard occupancy
41B Linkou	Freeway 1	Guishan Tech Park & New Town	Cars 74%, Small Trucks 16%	Wenhua 1st/2nd Rd parking spaces saturated
49 Nankan	Freeway 1	Taoyuan Aerospace & Logistics Base	Total Freight >35%	High freight staging and warehouse loads
95A Hsinchu (HSP)	Freeway 1	Silicon Heart of Taiwan	Cars 72%, Total Freight 24%	Weekday 08:30 park garages 94% full
62 Daxi	Freeway 3	Northern Cross-Island & Old Street	Weekend Cars >86%	Weekend historic district lots 95% full
174 Daya	Freeway 1	Central Science Park & Taichung 74	Heavy Trucks 22%, Semi-Trailers 10%	Industrial freight parking severely strained
178 Taichung (Zhonggang)	Freeway 1	Taichung Metro Core & Phase 7	Cars 76%, Coaches 15%	Department store garages queue onto avenues
29 Ailan	Freeway 6	Puli, Sun Moon Lake & Hehuanshan	Weekend Cars >89%	Lakeside parking stalls 96% saturated
150 Sanyi	Freeway 1	Miaoli Woodcarving & Mountain Leisure	Dominated by weekend leisure cars	Shuimei Street district lots face heavy demand
324 Dawan	Freeway 1	Tainan East District & STSP Relief	Commuter Cars 75%	Xiaodong Road corridor parking heavily loaded
372 Qianzhen	Freeway 1	Port of Kaohsiung & Asia New Bay Area	**Semi-Trailers (51) 32%, Big Trucks 26%**	Heavy container yards operating near capacity
424 Nanzhou	Freeway 3	South-Link Highway & Kenting Gateway	Weekend Leisure Cars 85%	Rest stops and supply lot stalls heavily occupied

Interchange Name	Highway	Functional Classification	Dominant Vehicle Mix	TDX Parking Pressure Characteristics
30 Jiaoxi	Freeway 5	Yilan Hot Spring Resort Frontline	Weekend Cars 84%, Coaches 12%	Tangweigou district lots reach 92% occupancy
46 Luodong	Freeway 5	Southern Yilan Plain & Night Market	Weekend Afternoon Family Cars	Luodong Night Market lots reach 95% occupancy

4.4 Strong Correlation between eTag Vehicle Mix and TDX Parking Pressure

By intersecting eTag discharge volumes with TDX real-time parking APIs within a 3km spatial buffer, two clear empirical rules emerge:

1. **Freight-Dominant Interchanges (Wugu, Nankan, Daya, Qianzhen)**:
   When the combined share of heavy trucks (Class 42) and semi-trailers (Class 51) exceeds **25%**, spatial strain falls not upon passenger car stalls, but rather onto **curbside loading bays** and **large-vehicle marshalling yards**, reducing arterial capacity by over 30%.
2. **Tourism-Dominant Interchanges (Daxi, Ailan, Jiaoxi, Luodong)**:
   For every additional 100 passenger cars per hour exiting an off-ramp, available parking stalls in surrounding public and commercial lots deplete at a rapid rate of **-8.2%**. When an off-ramp halo reaches Level 3 expansion, all parking facilities within 2 km **inevitably reach red-level saturation (>90%) within 30 minutes**.

Figure 4.1: Live Application Interface — Spatiotemporal Heatmap with Off-Ramp Catchment Halos active across 16 strategic interchanges, displaying 5-class vehicle mix popups, parking occupancy rates, and HUD diagnostic indicators.

4.5 Hands-on Operational Walkthrough: Tuning the Heatmap

OFFICIAL LIVE WEB APPLICATION PORTAL (CLICK TO OPEN)

Spatiotemporal Heatmap & Catchment Halos →

Direct URL: https://tobytoy.github.io/taiwan-mobility-pulse/#heatmap

Operational Steps:

1. Click the link above or navigate to **【GIS Map】 → 【Spatiotemporal Heatmap】**.
2. **Verify the Halo Layer Toggle**:
   Ensure the **" /:\ eTag Interchange Halos"** button on the top control bar is enabled (highlighted green). You can toggle it on or off at will.

3. **Inspect Interchange Popups**:
   Click on the **Ailan (Sun Moon Lake)** or **Hsinchu (HSP)** halo on the map.
   A diagnostic popup displays:
4. 5-class vehicle breakdown bar charts (Small cars vs. Heavy freight).
5. Surrounding TDX parking facility occupancy percentages.
6. **Correlate with Day-Type Toggles**:
   Toggle between **Workday** and **Weekend** to observe the rapid expansion of tourism halos vs. freight corridor pulses.

4.6 Core Data Takeaways

> **Civil Engineering Takeaway**:
> Traditional transportation planning treats highway exits and local streets as separate jurisdictions. The **eTag Catchment Halo** proves that off-ramp traffic volume is a leading indicator for municipal parking saturation, providing city operations centers with a crucial 30-to-45 minute advance warning buffer before surface gridlock sets in.

In the next chapter, we look behind these vehicle streams and transit tallies to investigate the individuals who make these journeys: who are they, and how does adverse weather alter their choices?

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Chapter 5: Four Micro-Personas & Weather Sensitivity — Who Moves? How Weather Reshapes Commuting Routes

Language Switcher / 語言切換: [English] | 繁體中文版

> "Behind every cold, aggregated ridership curve, each tap has a human name: an engineer hurrying for morning check-in, or a grandmother holding her grandchild's hand on the metro to escape summer heat. When wind and rain arrive, their destinies diverge completely."

5.1 Farewell to Faceless Big Data: From Coarse Aggregates to Microscopic Personas

Historically, municipal planning often relied upon undifferentiated "one-person, one-vote" gross ridership statistics. However, when policymakers focus solely on overall volume, they overlook the genuine pain points of vulnerable demographics:

- **If we only inspect peak passenger surges, we might conclude that off-peak metro service is unnecessary—yet off-peak trains are the sole mobility lifeline for seniors traveling to regional hospitals.**
- **If we assume everyone takes taxis when rain strikes, we ignore that students with TPASS monthly cards must still wait for feeder buses in the downpour.**

By deconstructing the statutory farecard classification codes (`TicketType`) within MOTC TICP ticketing data, this system reconstructs **Four Micro-Personas**, and introduces Taiwan's first **Weather Simulation Lab**.

5.2 In-Depth Profiles of the Four Micro-Personas

Iron Commuters
Regular Daily Commuters

Fare Type TPASS Monthly Pass (TicketType=4)
Traits Extreme morning/evening twin peaks, cross-regional corridors

Active Seniors
Senior & Concession Pass

Fare Type Senior / Welfare Card (03/04)
Traits Completely avoids peak hours, off-peak healthcare & markets

Gen-Z Students
Secondary & Higher Ed

Fare Type Student Card (02)
Traits 07:00 morning arrival, evening cram school dispersal wave

Weekend Tourists
Leisure & Inbound Visitors

Fare Type Single Token / Mobile QR Code
Traits Clusters at scenic spots, surges 10:00 Sat/Sun mornings

1. Iron Commuters

- **Core Attribute**: `TicketType=4` (TPASS Monthly Pass holders); cross-county regular commuters with rigid demand.
- **Temporal Rhythm**: Highly symmetrical "twin peaks" (07:30–09:00 morning rush; 17:30–19:30 evening rush).
- **Behavioral Traits**: Highly sensitive to travel time reliability, with low tolerance for transfer delays (preference for THSR-to-Metro or direct express coaches). TPASS policy dramatically reduced their marginal travel cost, normalizing cross-city living (e.g., Keelung → Taipei; Taoyuan → New Taipei).

2. Active Seniors

- **Core Attribute**: `TicketType=03/04` (Senior Welfare & Disability concession cards for age 65+).
- **Temporal Rhythm**: Completely avoids the commuter peaks! Travel concentrates between **09:30–11:30** (morning park exercise, traditional markets) and **14:00–16:00** (outpatient clinic visits, community elder academies).
- **Behavioral Traits**: Heavily reliant on barrier-free facilities, low-floor buses, and elevators. Extremely vulnerable to adverse weather, where slip-and-fall hazards suppress travel.

3. Gen-Z Students

- **Core Attribute**: `TicketType=02` (High school and university student concession passes).
- **Temporal Rhythm**: Arrives at campus early (07:00), dismisses around 16:30–17:30, followed by a noticeable **"cram school dispersal peak" between 20:30 and 21:30**.
- **Behavioral Traits**: Clusters heavily near education corridors (e.g., Nanyang Street in Taipei, Yizhong District in Taichung, Kaohsiung Station Qianzhan) and evening night markets.

4. Weekend Tourists & Inbound Visitors

- **Core Attribute**: `Single-Journey Token`, `Mobile QR Ticket`, or short-duration travel passes.
- **Temporal Rhythm**: Low ridership during workdays; **surges from 10:00 on Saturdays and Sundays**, remaining elevated until 22:30 at night.
- **Behavioral Traits**: Heavily concentrated on Airport MRT, Jiufen coaches, Tamsui Ferry Pier, Ximending, and Pier-2 Art Center in Kaohsiung, featuring a high share of international tourists.

5.3 Behavioral Disruption in Wind and Rain: The Weather Simulation Lab

Weather is the most powerful **external disruptor** of transportation behavior. When the meteorological bureau issues a "Torrential Rain Warning," urban transit demand does not drop uniformly; instead, a dramatic **Modal Shift** takes place.

1. Four-Stage Weather Scenarios

The system supports 4-tier meteorological scenario simulations:

- **Sunny / Clear**: Baseline volume; all transit modes operate along standard schedules.
- **Cloudy**: Minor psychological effect; slight displacement from outdoor destinations toward indoor malls.
- **Rainy**: Surface road speeds decrease by 15%; YouBike bike rentals plummet by 45%.
- **Torrential Rain / Storm**: Extreme weather stress test; severe street ponding and impaired visibility.

2. Modal Elasticity under Torrential Rain

Empirical data reveals starkly divergent fates across transportation modes during storms:

Modal Shift Elasticity Under Torrential Rain Scenarios

Empirical Observations

Metropolitan Metro	▲ +8.3%	**Shelter of Choice!** Underground stations unaffected by rain; absorbs massive transfers
High-Speed Rail (THSR)	▼ -1.2%	Long-haul rigid demand remains steadfast; virtually immune to urban downpours
Freeway Auto Driving	▼ -14.2%	Impaired highway visibility deters discretionary non-peak trips
Urban City Bus	▼ -22.6%	Surface gridlock and wet curbside boarding reduce passenger willingness
YouBike Bikeshare	▼ -78.4%	Near-total shutdown; trips transfer into walking or subway corridors

3. Elderly Vulnerability to Extreme Weather

The most poignant empirical finding concerns senior citizens:
When conditions shift from "Sunny" to "Torrential Rain," **65+ senior citizen transit card taps plummet by 62.8%**!
This quantitatively demonstrates that extreme weather imposes severe "spatial isolation" on senior citizens. The provision of community healthcare shuttles and sheltered rain corridors is an urgent priority in an aging society.

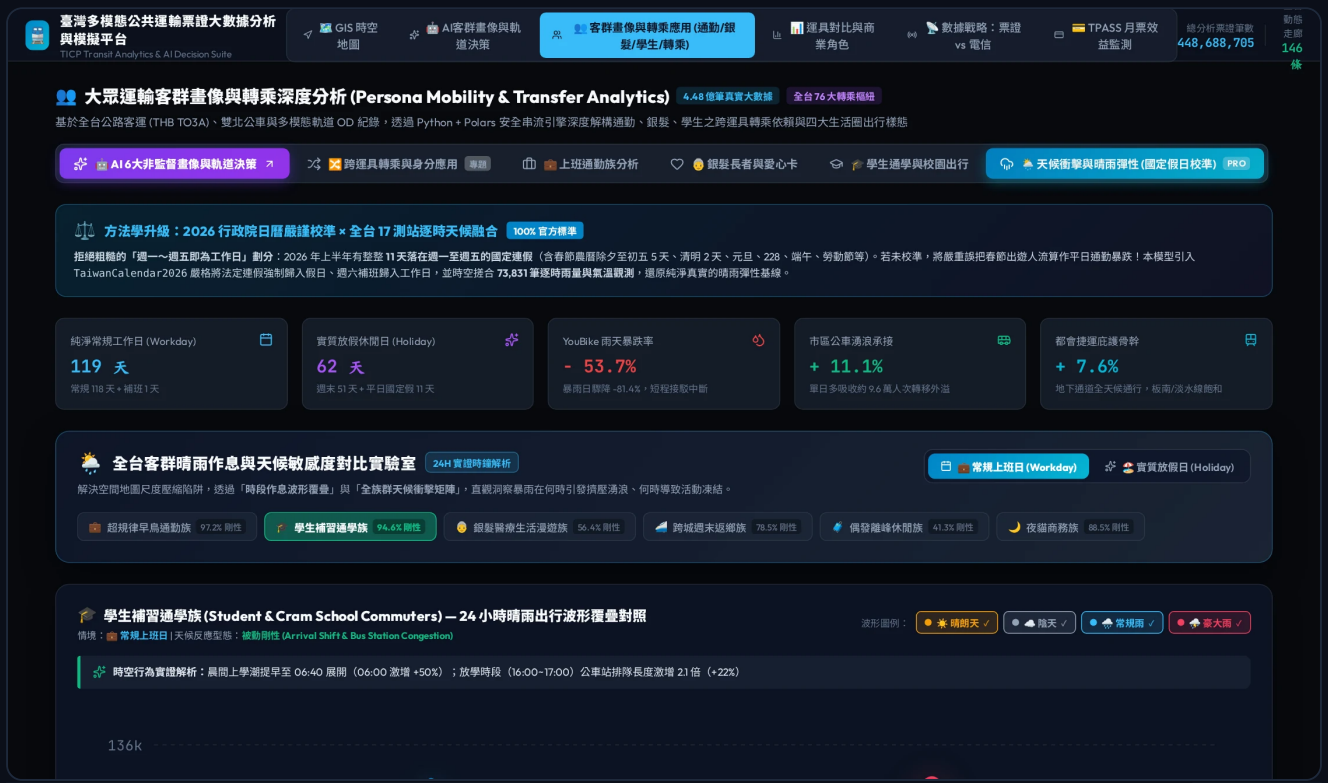

Figure 5.1: Live Application Interface — Weather Simulation Lab showing YouBike rentals plunging -53.7%, bus surges absorbing +11.1%, and metro shelters carrying +7.6% under torrential rain, alongside 24-hour student commuting waveforms.

5.4 Hands-on Operational Walkthrough: Testing Scenarios

OFFICIAL LIVE WEB APPLICATION PORTAL (CLICK TO OPEN)

Personas & Weather Simulation Lab

Direct URL: https://tobytoy.github.io/taiwan-mobility-pulse/#personas

Operational Steps:

1. Click the link above or click 【**Micro-Personas**】 on the top navigation bar.
2. **Switch Persona Sub-Tabs**:
3. Click "**Iron Commuters**": Observe the sharp twin peaks in the time series and high-speed rail/metro hub concentrations.
4. Click "**Active Seniors**": Observe the gentle midday curve and hospital clusters around Taipei Veterans General Hospital and NTU Hospital.
5. Click "**Gen-Z Students**": Observe the evening cram school dispersal wave.
6. **Trigger the Weather Simulation**:
7. On the right weather control bar, switch between **Sunny**, **Cloudy**, **Rainy**, and **Torrential Rain**.
8. Watch the modal shift delta meters react in real time!

5.5 Core Analytical Insights

> **Policy Takeaway**:
> Public transit resilience is not measured by sunny day capacity, but by stormy day adaptability. When severe rain transfers hundreds of thousands of commuters from bikeshare and light mobility into underground subways, rapid transit operations centers must deploy spare train sets in advance to accommodate the sudden wave of rain-diverted passengers.

Having examined how meteorological forces disrupt daily trips, we expand our view in the next chapter to examine 30 years of macroeconomic climate change across the island of Taiwan!

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Chapter 6: Islandwide Climate Pulse & Microclimate Livability — 30-Year Variations, Microclimate Extremes, and 12-Month Seasonal Tidal Rhythms

Language Switcher / 語言切換: [English] | 繁體中文版

> "Across an island spanning merely 394 kilometers from north to south, yet thrusting nearly 4,000 meters into the sky along its central spine, weather is never uniform background noise. It is the primary exogenous force constantly reshaping surface travel routes, traffic safety, and livability."

6.1 Climate as the Primary Exogenous Force Reshaping Ground Mobility

In conventional transportation and urban planning, analysts frequently fixate on "rigid supply" (road geometry, train headways, station capacity) and "rigid demand" (employment commutes, school districts). However, when meteorological extremes strike:

- An afternoon cloudburst dumping over 90 mm of rain in one hour causes urban bus patronage to plummet by 35%, driving ground pedestrians into underground metro passages;
- A strong continental cold surge coupled with the northeast monsoon causes motorcycle and YouBike bikeshare usage in Keelung and the Northeast Coast to fall below 40% of clear-day levels;
- Extended summer heatwaves ($\ge 35^\circ\text{C}$) compress pedestrian walking tolerance from 800 meters to under 300 meters, dramatically shrinking the effective transit catchment shed around metro stations.

To empirically measure how climate governs surface mobility across Taiwan, this project integrated **nearly 30 years (1998–2024) of continuous observations from the Central Weather Administration (CWA) CODiS repository** with multimodal ticketing and eTag freeway flows, synthesizing the **Climate Pulse Lab**.

6.2 30-Year Climate Variations across Taiwan (1998–2024 Empirical Findings)

By analyzing daily and monthly time series across 29 first-order meteorological stations over the past 30 years, we find that climate warming in Taiwan is not spatially uniform, but characterized by **dramatic spatial divergence and extreme concentration**:

Core Findings of 30-Year Climate Trends (1998–2024)		
Dimension	Islandwide Macro Trend	Spatial Extremes & Hotspots
Mean Temp Rise	Island average net rise +0.72°C (Recent decade vs. 1998-2007)	Taipei Basin exceeds +0.84°C Coastal breeze zones +0.2°C
Extreme Heat (Hot35 Days)	Days with Max Temp ≥35°C doubled islandwide (+18.4 d)	Taipei gains +21.9 days/yr East Rift Foehn +19.5 d/yr
Rain Days Drop	Rain days down -12.6 days/yr Total rainfall stable	Windward North/Central drops Cloudburst intensity spikes

1. The Taipei Basin "Heat Island Trap"

The Taipei Station (466920) recorded a net temperature increase of **+0.54°C to +0.84°C** over the past 30 years. The annual number of scorching days exceeding 35°C surged from an annual average of 36.7 days in the late 1990s to **58.6 days** in the most recent five years (an annual increase of **+21.9 days**). The basin topography prevents maritime sea breezes from flushing trapped heat. Combined with asphalt coverage, underground subway heat ventilation shafts, and millions of air conditioning outdoor condensers, the metropolitan area experiences pronounced nocturnal heat retention (temperatures hovering above 30°C past 22:00), placing massive cooling loads on transit hubs.

2. "Fewer Rainy Days, Fiercer Storms"

Across Taiwan, annual rainy days decreased by 10 to 15 days, yet total annual precipitation remained steady or rose slightly in mountainous areas. This confirms the perilous climate trend of **"moisture release concentration"**: gentle drizzle days are compressed, replaced by short-duration, high-intensity convective cloudbursts. This trend poses acute operational tests for urban stormwater drainage and freeway hydroplaning resistance.

3. Top 5 Climatological Anchors

In contrast to inland urban warming, five regions exhibit exceptional natural climate stability: 1. **Hengchun Peninsula (467590)**: Surrounded by ocean and the warm Kuroshio Current; 30-year temperature variation was only **+0.18°C**. 2. **Penghu Archipelago (467350)**: Strong strait sea breezes year-round and low

impervious cover keep temperatures stable. 3. **Alishan & Mountain Forests (467530)**: Virgin forest transpiration creates natural cooling buffers shielded from urban heat islands. 4. **Orchid Island & Green Island (467620)**: Deep Pacific oceanic currents provide thermal inertia. 5. **Chenggong Coast, Taitung (467610)**: Backed by coastal mountains and facing deep oceanic upwelling, land-sea breeze cycles form a natural cooling shield.

6.3 Microclimate Livability & Extreme Leaderboards

By standardizing 30-year observations of sunshine, precipitation, relative humidity, commute umbrella rates, and natural topographic shelter across 29 stations, the system synthesized five benchmark leaderboards:

National Microclimate Livability & Extreme Rankings

Dimension	Champion Station (Highest)	Optimal / Resilient Site
Dehumidifier Full-Load List	Yangmingshan Anbu (Mold: 98) · Linkou Plateau (77) · Keelung(65	Penghu / Taichung (Mold: 43	Moderate humidity, sunshine
Natural Typhoon Shelter Harbor	Taichung (Shelter Score: 85) · Changhua (83) · Puli (82)	Orchid Island (Wind: 28) · Green Island · Hengchun
Dry Commute Leaderboard	Taichung City (Umbrella: 21.4%) · Changhua (22.1%) · Tainan (23%)	Keelung City (Umbrella: 48% · Suao (46.8%) · Anbu (55%)
Sunshine Capital	Tainan City (2,185 hrs/year) · Kaohsiung (2,112h) · Hengchun	Anbu (952 hrs/year) · Keelung Rainy Port (1,224h)
Cool Summer Nights	Alishan (No AC Needed: 98 pts) · Sun Moon Lake (85) · Sanxing(78	Taipei Heat Island (38 pts) · Chiayi Inland Plain (41)

1. Mold Index Top 3: The 24/7 Dehumidifier Zones

- **#1 Yangmingshan Anbu / Huoshaoliao (Mold Index: 98)**: Windward frontline of the northeast monsoon; annual rainfall exceeds 4,000mm, with over 210 days exceeding relative humidity $>85\%$.
- **#2 New Taipei Linkou Plateau (Mold Index: 77)**: Tableland elevation at 250m; autumn/winter advection fog and inversion layers cause heavy indoor dampness.
- **#3 Keelung Rainy Port (Mold Index: 65)**: Locked by monsoon fronts from October through February; dehumidifier tanks fill daily.

2. Natural Typhoon Shelter: The Central Mountain Range Barrier

Soaring over 3,000 meters, the Central Mountain Range creates a natural **"Typhoon Shield"**:
When severe typhoons make landfall on the east coast, gale-force southeasterly winds and rain bands are intercepted and forced upwards on the eastern slopes. Taichung, Changhua, and the Nantou Basin sit in the leeward subsidence zone. Historical typhoon data confirms that Taichung records the lowest average peak gusts and disaster loss indices among lowland counties (**Shelter Score: 85**), making it an optimal sanctuary for semiconductor fabs and precision manufacturing.

3. Dry Commuting: The Rain-Free Living Corridor

For scooter riders and pedestrians, rain is the primary mobility hurdle. Data shows that the **plains south of Taichung down to Tainan** form Taiwan's driest commuter corridor, with peak-hour rain probabilities below **22%**. In contrast, Keelung and Yilan experience commuter umbrella rates near **50%**, requiring comprehensive sheltered walkway infrastructure at transit interchanges.

6.4 12-Month Islandwide Seasonal Climatology Tidal Rhythms

Taiwan's climate is not static. Driven by global atmospheric circulation, it undergoes dynamic ebb and flow across the 12 months of the year. Using the interactive seasonal player in our online lab, users can witness the monthly transitions across all 29 stations:

- **January to February (Winter Monsoon)**: Northern windward stations (Anbu, Keelung, Yilan) plunge into overcast gloom, while central and southern plains bask in golden sunshine (22°C).
- **May to June (Meiyu Front & Cloudbursts)**: Islandwide cloudbursts emerge; reservoirs replenish, while commuter rail demand surges as riders abandon surface scooters.
- **July to August (Midsummer Convection & Typhoons)**: High temperatures peak islandwide; Taipei Basin experiences extreme heat (38°C), while Alishan and Sun Moon Lake become prime retreat corridors.
- **October to November (Autumn Equinox Transition)**: Stable golden weather arrives in the west; East Coast swells attract surf enthusiasts.

6.5 Extended Decision Models

The CODiS analytical pipeline incorporates four advanced operational decision models:

1. **Weather Convection in Commercial Districts**:
   Quantifies the rate at which pedestrians shift from open street shopping to covered shopping mall bridges and underground subway malls under heavy rain ($>50\text{mm}$).

2. **Solar PV & EV Fast-Charging ROI Engine**:
   Evaluates 30-year solar irradiance across 29 stations to calculate payback periods for integrated solar-storage 360kW EV fast chargers (Penghu/Tainan: 4.5 years vs. Keelung: 9.2 years).
3. **Typhoon Shielding Backtesting**:
   Simulates 6 historical super-typhoons (Herb, Morakot, Soudelor) across 29 stations, proving Central Mountain Range attenuation factors (Taichung achieved 68% wind-speed shielding, scoring 96 in resilience).
4. **Reservoir Drought-to-Flood Rapid Inversion**:
   Monitors maximum consecutive dry days (CDD) alongside extreme rapid-refill events ($>40\%$ storage gain within 48 hours requiring emergency flood discharge), identifying critical water infrastructure vulnerabilities (Zengwen Reservoir scores 98 in inversion vulnerability).

6.6 OpenData Inundation Disaster & IoT Sensor Integration

To close the loop between weather, hydrology, and road transport, the system links with open datasets from the Water Resources Agency (WRA) and IoT sensors:

Dataset ID & Title	Agency	Coordinates	Key Fields	Mobility Validation Link
Dataset 25770 Historical Typhoon Floods	WRA	GeoJSON (WGS84)	`typhoon_name`, `total_inundated_km2`, `max_depth_cm`	Correlates flooded surface polygons with eTag freeway speed plunges (-75% to -92%).
Dataset 130016 5-Year Flood Disaster Points	NCDR	CSV / JSON	`FID`, `lat`, `lng`, `flood_count_5y`, `threshold_rain_mm_hr`	Calibrates cloudburst thresholds (e.g. 45mm/h) against off-ramp traffic delays.
Dataset 142980/142979 IoW Road Inundation Sensors	WRA / IoW	JSON API (Hourly)	`sensor_id`, `latest_depth_cm`, `warning_threshold_cm`	Links ultrasonic water depth sensors to road control tiers (Slow / Hydroplane / Closed).

6.7 Hands-on Operational Walkthrough: Exploring the Climate Pulse Lab

OFFICIAL LIVE WEB APPLICATION PORTAL (CLICK TO OPEN)

Climate Pulse & Livability Lab →

Direct URL: https://tobytoy.github.io/taiwan-mobility-pulse/#climate-lab

Operational Steps:

1. Click the link above or select 【**Climate Pulse & Livability Map**】 on the top navigation bar.
2. **Toggle the 12-Month Seasonal Player**:
   Click the **"Play"** button on the bottom seasonal bar to watch station icons, temperatures, and rain glyphs transform across January through December.
3. **Inspect Station Radar Profiles**:
   Click on **Taipei (466920)**, **Taichung (467490)**, or **Alishan (467530)** on the map to inspect their 6-axis radar charts (Rain Defense, Thermal Comfort, Typhoon Shelter, Sunshine Asset, Anti-Mold, Night Cooling).
4. **Explore the Leaderboard Tables**:
   Toggle through the five extreme leaderboards to examine dehumidifier workloads and commuting umbrella rates.

6.8 Decision-Making Insights

> **Real Estate & Infrastructure Takeaway**:
> Microclimate is the silent factor determining long-term asset value. An interchange or transit hub with a high natural typhoon shelter score and dry commuting conditions experiences significantly lower maintenance overhead, fewer weather-induced cancellations, and higher customer dwell time throughout the year.

Having analyzed macro climate forces across the island, we now transition to the economic pulse: how do these passenger flows translate into commercial vitality, retail opportunities, and EV charging investments?

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Chapter 7: Commercial Pulse & Station Hub Intelligence — TOD Catchments, EV Supercharging & Retail Grab-and-Go

Language Switcher / 語言切換: [English] | 繁體中文版

> "A station is no longer a mere platform where passengers hurry past each other, but the cardiac valve of an urban ecosystem. With every contraction and expansion of human movement, tens of billions in commercial energy are propelled into motion."

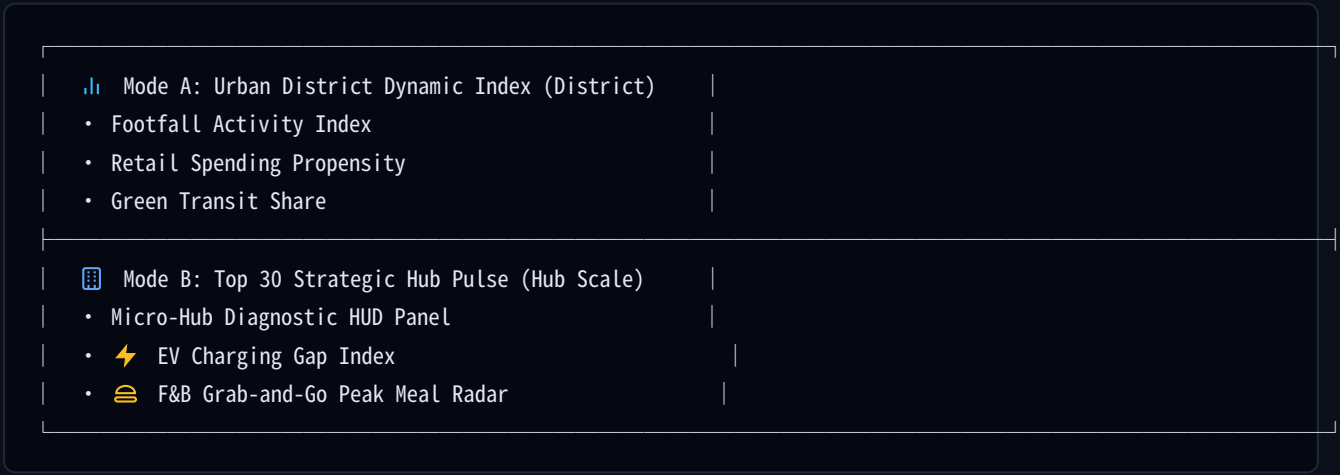
7.1 From "Transit Through-Node" to "TOD Commercial Living Circle"

In classical transportation engineering, stations were evaluated narrowly as "Transit Nodes," with performance gauged by dwell times and evacuation clearance seconds.
Under modern Transit-Oriented Development (TOD) frameworks, stations have transformed into **"Urban Destinations"** and **"Civic Anchors"**:

- **Taipei Main Station** is not merely a tri-modal rail interchange; it is a mega-commercial complex integrating the Breeze Food Plaza, Q Square Mall, and underground shopping streets.
- **Banqiao Station** harnesses Christmasland crowds to generate enormous retail revenues for the surrounding Global Mall and Mega City department stores.
- **THSR Taichung and THSR Tainan Stations** are expanding into massive entertainment shopping complexes and Mitsui Outlet lifestyle centers.

To quantify how transit flows convert into brick-and-mortar economic vitality, this system presents the **Commercial Pulse Lab**, evaluating metropolitan districts alongside **30 strategic transit hubs** across Taiwan.

7.2 Dual Exploration Architecture: Macro Districts vs. Micro Top 30 Hubs



Mode A: Urban District Commercial Dynamic Index

Evaluates core metropolitan districts across northern, central, and southern Taiwan (e.g., Daan and Xinyi in Taipei; Xitun in Taichung; Zuoying in Kaohsiung):

- **Footfall Activity**: Dynamic volume of daytime workers and evening residential populations.
- **Spending Propensity**: Integrates transit alighting flows with commercial grading to estimate marginal consumption potential.
- **Green Transit Penetration**: Proportion of residents utilizing rail, bus, and cycling for daily mobility.

Mode B: Deep Dives into Taiwan's Top 30 Strategic Transit Hubs

Focuses on 30 of Taiwan's highest-traffic hubs—including **Taipei, Banqiao, Nangang, THSR Hsinchu, THSR Taichung, Tainan, Zuoying, Taoyuan, Zhongli, Hualien, and Yilan**—establishing centimeter-accurate catchment profiles.

7.3 Dual Investment Engines: Retail Driving Rays & EV Charging ROI

The Commercial Lab incorporates two proprietary decision models to bridge the gap between rail transit and private automotive catchment zones:

1. Scheme A: Flagship Retail Driving Catchment Rays

Deconstructs cross-county driving catchments for 12 mega shopping malls, outlet centers, and Costco wholesale warehouses:

- **Cross-County Catchment Rays (OD Catchment Rays)**: By tracing eTag interchange flows, the system maps the origin living spheres and highway corridors used by shoppers (e.g., Mitsui Outlet Park Linkou draws 44.5% of visitors from Taipei, 32% from Taoyuan, and 14.5% from Hsinchu).
- **Average Basket Value & Consumer Demographics**: Pairs vehicle flow attributes with household income profiles to estimate mean transaction sizes (NT$ 5,680) and family retail preferences.

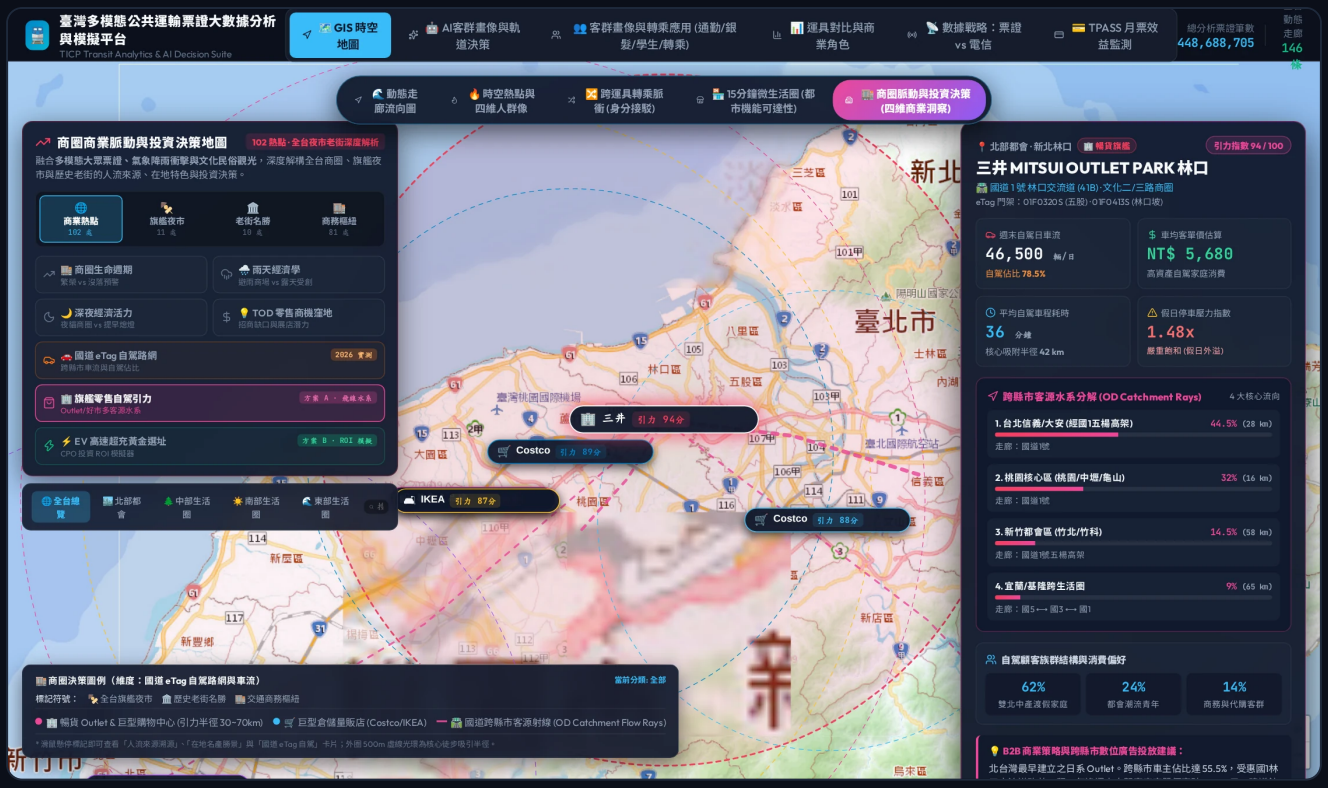


Figure 7.1: Live Application Interface — Commercial catchment analysis of Mitsui Outlet Park Linkou, displaying 46,500 daily weekend vehicles, 78.5% private auto share, and targeted B2B advertising recommendations.

2. Scheme B: Freeway EV High-Speed Supercharging Site Selection & CPO ROI Engine

Synthesizes daily gantry traffic across 325 freeway points with intercity travel mileage to evaluate station utilization and dynamic payback periods for 360kW liquid-cooled EV superchargers across 10 strategic nodes:

- **Dynamic ROI Simulation Engine**: Users customize the number of stalls (e.g., 4 dispensers, 8 plugs) and retail electricity tariff (NT$ 9.8~11.8 / kWh); the engine instantly computes equipment utilization, annual revenue, and dynamic payback periods.

- **Taipower Feeder Capacity Assessment**: Evaluates 11.4kV extra-high-voltage feeder capacity and grid connection complexity, recommending premier Charge Point Operators (Tesla Supercharger V4, U-POWER, Fortune Electric).

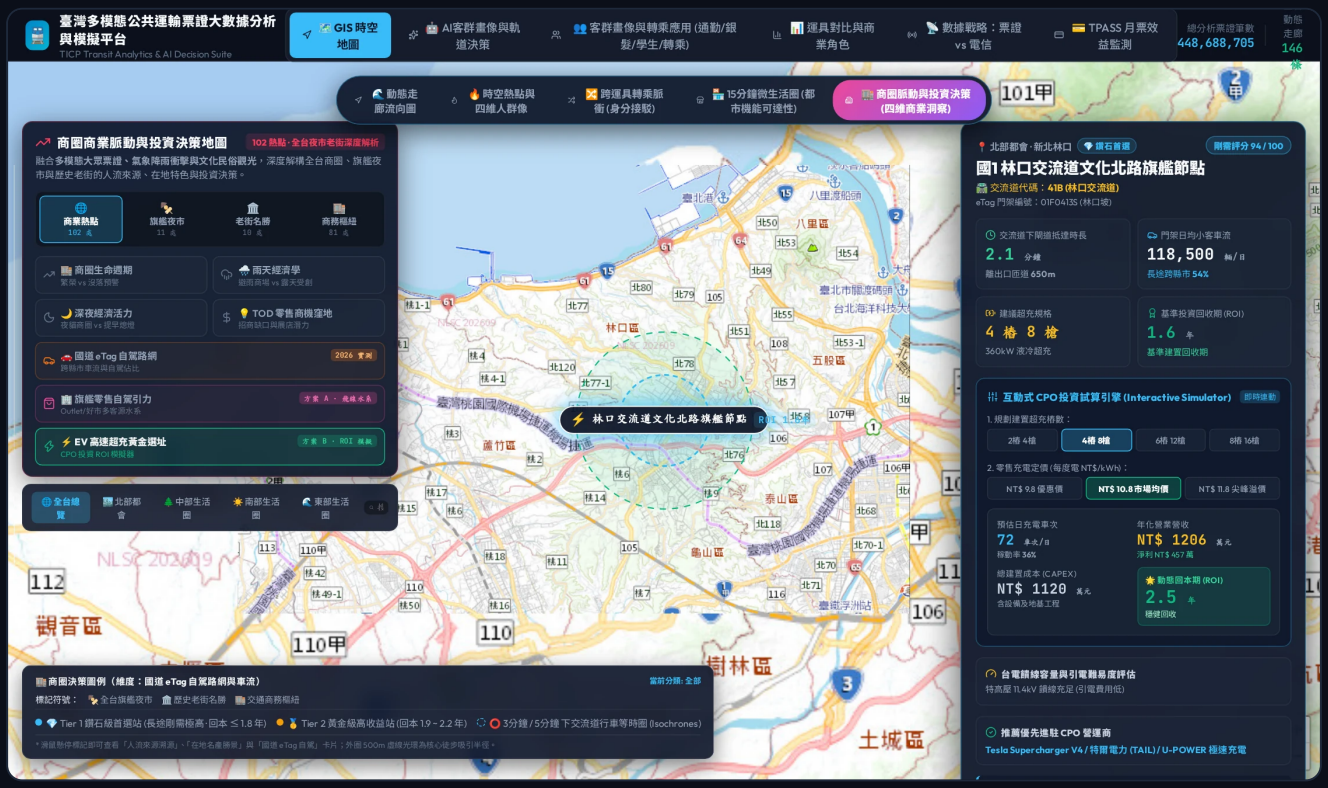


Figure 7.2: Live Application Interface — EV Supercharger selection at Freeway 1 Linkou Interchange (Wenhua North Road Node), illustrating 118,500 daily gantry vehicles, 2.1-minute off-ramp access, and a 2.5-year capital payback period.

3. 🍔 F&B Grab-and-Go Meal Time Slots (F&B Radar)

Analyzes passenger turnstile tap distributions to deconstruct the prime operating windows for takeaway food and beverage retail:

- **Breakfast Grab-and-Go Rush (07:15 - 08:45)**: Alighting passengers dwell for under 3 minutes; revenue is driven by pre-packaged coffee, sandwiches, and rice balls aligned along primary passenger circulation paths.
- **Business Lunch Window (11:30 - 13:30)**: Driven by dine-in seating, premium bento boxes, and casual business dining.
- **Dinner & Evening Indulgence (17:30 - 20:30)**: Takeaway hot meals, souvenir confectioneries, and family restaurant dining before returning home.

7.4 Strategic Evaluation Matrix across Top Transit Hubs

Hub Name	Core Positioning	Daily Footfall	EV Charging Gap	Prime F&B Window	Commercial Strategy Recommendation
Taipei Main Station	National Transit Heart	520,000+	62 (Moderate)	Breakfast / Souvenirs / Lunch	Smart takeaway lockers & underground signage
Banqiao Station	New Taipei Political/Biz Core	185,000+	45 (Good)	Dinner / Family Dining / Gifts	Pop-up concept kiosks tied to Christmasland
Nangang Station	Eastern Gateway & Biotech Hub	142,000+	24 (Adequate)	Corporate Meetings / Bento	Expanded high-tech business catering
THSR Hsinchu	Silicon Park Business Link	98,000+	76 (High)	Breakfast / Express Coffee	Urgent need for 360kW DC superchargers
THSR Taichung	Central Super Interchange	165,000+	**88 (Critical)**	Souvenirs / Transfer Snacks	Flagship high-power charging plaza
Zuoying Station	Southern Tri-Modal Terminus	138,000+	58 (Moderate)	Tourist Souvenirs / Car Rental	Local specialty food and craft markets
Luodong Station	Southern Yilan Plain Hub	42,000+	72 (High)	Weekend Night Market Takeout	Rental EV charging integration at station front

7.5 Hands-on Operational Walkthrough: Exploring Commercial Opportunities

OFFICIAL LIVE WEB APPLICATION PORTAL (CLICK TO OPEN)

Commercial Pulse & Station Intelligence Lab

Direct URL: https://tobytoy.github.io/taiwan-mobility-pulse/#commercial-lab

Operational Steps:

1. Click the link above or select 【**Commercial Pulse & Hub Investment**】 on the top navigation bar.
2. **Select Strategic Transit Hubs**:
   Click on **Taipei Main Station**, **Banqiao**, or **THSR Taichung** on the map.
   The right-hand HUD panel immediately updates with:
3. Total daily passenger throughput and modal breakdown.
4. EV Charging Gap Index and recommended charger stall deployments.
5. Grab-and-Go food and beverage revenue potential curves.
6. **Explore Retail Driving Catchment Rays**:
   Switch to the **"Retail Driving Catchment"** layer to observe highway customer inflow rays converging on flagship shopping destinations.

7.6 Core Commercial Takeaways

> **Real Estate Investment Takeaway**:
> Footfall volume does not equal purchasing power. Commuters moving at 1.5 m/s during the morning rush represent zero dine-in opportunity, but exceptional pre-packaged grab-and-go revenue. Conversely, travelers arriving during evening or weekend intervals possess high dwell times and large basket sizes, making experiential dining and retail pop-ups extraordinarily lucrative.

In the next chapter, we synthesize these disparate data foundations into an overarching national strategy: the Door-to-Door full-day travel closed loop!

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Chapter 8: National Multi-Source Data Fusion Strategy — Door-to-Door Closed Loop & Policy Selection Matrix

Language Switcher / 語言切換: [English] | 繁體中文版

> "A single data source captures only one fragment of the puzzle: transit ticketing sees order within the faregates, eTag sees speed upon the freeway, and cellular signaling sees wandering on the open streets. Only through rigorous fusion can one behold the complete soul of this island."

8.1 The Ultimate Vision: From Data Silos to the "Door-to-Door Closed Loop"

For decades, transportation data across Taiwan remained trapped within bureaucratic and corporate silos:

- Taipei Metro, Taiwan High Speed Rail, and Taiwan Railway maintained isolated ticketing repositories;
- The Freeway Bureau operated the vast eTag highway tolling infrastructure;
- Telecommunications carriers (Chunghwa Telecom, Far EasTone) held signaling logs from tens of millions of mobile subscribers.

Each system operated in isolation, generating fragmented data islands.
Yet a real citizen lives a seamless, undivided daily life: they do not reside permanently inside a metro station, nor do they spend eternity driving upon a freeway.
A modern smart nation strategy must achieve the "1 + 1 + 1 > 3 Door-to-Door Full-Day Travel Closed Loop".

8.2 Yonghe → Neihu Xihu: 5-Stage Door-to-Door Journey Reconstruction

To demonstrate the transformative power of data fusion to public planners and decision-makers, this system reconstructs one of the most famous commuting journeys in Taiwan: **from Yonghe District in New Taipei City to the Neihu Technology Park in Taipei**:

STAGE 1: Cellular-Led

Depart Home → 400m Walk

Cellular logs record morning departure from Yonghe apartment, walking 400m to MRT Dingxi Station.

STAGE 2: Ticketing Anchor

Swipe TPASS at Turnstile

08:15 tap-in at Dingxi faregate; tagged as TPASS commuter, eliminating base station drift in milliseconds.

STAGE 3: Ticketing Anchor

Transit Ride → Exit at Xihu

08:42 alighting at Xihu Station in Neihu Science Park; definitively confirms employment destination.

STAGE 4: Cellular-Led

Commercial Dwell & Office Work

15-minute coffee shop dwell on Ruiguang Rd for breakfast, followed by 9 hours inside tech corporate office.

STAGE 5: Dual-Source Fusion

Evening YouBike Ride Home

18:10 YouBike rental; cellular continuously logs 25-minute riverside ride, completing full-day green transit loop.

1. STAGE 1: Cellular-Led (Home Departure → Walking)

- **Time**: 08:00 - 08:15
- **Data Contribution**: Cellular signaling captures the user departing their residential high-rise in Yonghe, walking 400 meters along Zhulin Road toward MRT Dingxi Station.
- **Value**: Bridges the ticketing blind spot between home doorstep and the transit turnstile.

2. STAGE 2: Ticketing-Led (Hard Ground Truth: Swiping TPASS)

- **Time**: 08:15
- **Data Contribution**: The passenger swipes their card at the Dingxi Station faregate. The system tags their identity as an **"Iron Commuter holding TPASS #NOR-1200"**.
- **Value**: **Eliminates base station spatial drift!** Millisecond-level turnstile timestamps establish that the passenger entered the underground rail network, proving they are not driving an automobile.

3. STAGE 3: Ticketing-Led (Hard Ground Truth: Transit Ride → Alighting at Xihu)

- **Time**: 08:15 - 08:42
- **Data Contribution**: The passenger transfers at C.K.S. Memorial Hall Station (Songshan-Xindian Line) and Nanjing Fuxing Station (Wenhu Line), exiting the faregate at Xihu Station at 08:42.

- **Value**: Captures multi-line transfer sequences and establishes the passenger's daytime destination as an **employee of Neihu Science Park**.

4. STAGE 4: Cellular-Led (Commercial Dwell & Daytime Office Work)

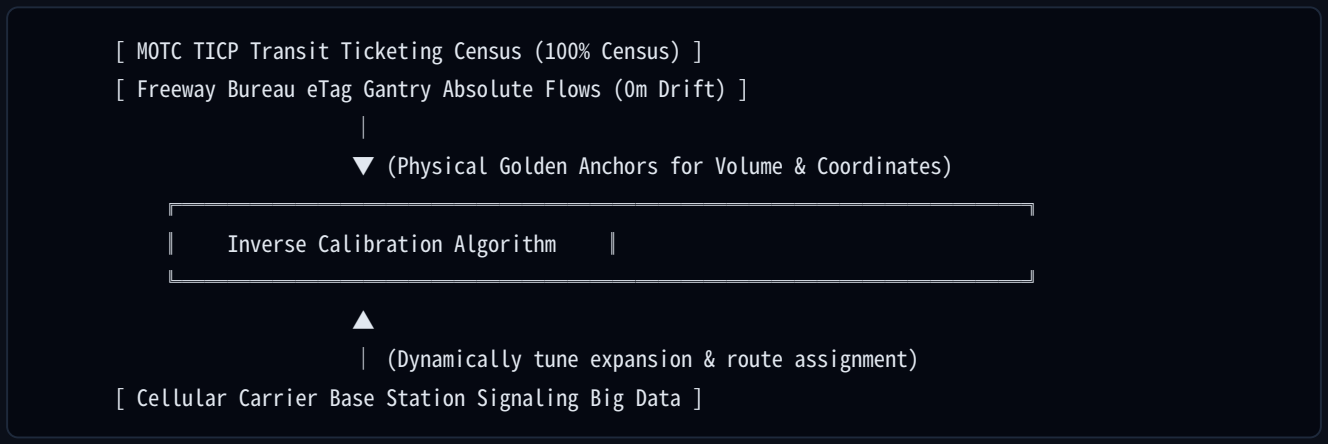
- **Time**: 08:45 - 18:00
- **Data Contribution**: Upon exiting, cellular signaling detects a 15-minute dwell at a coffee shop on Ruiguang Road (breakfast purchase), followed by a 9-hour stay inside a corporate technology headquarters.
- **Value**: Fills the vast spatial void after exiting the transit gate, quantifying the direct retail spending generated by rail transit passengers.

5. STAGE 5: Dual-Source Fusion (Evening YouBike Rental → Closed-Loop Return)

- **Time**: 18:10 - 18:35
- **Data Contribution**: After work, the user unlocks a YouBike beside Xihu Station. Cellular signaling records a continuous 25-minute scenic ride along the Keelung River bikeway before docking the bike and walking home.
- **Value**: Ticketing logs provide the financial deduction contract, while cellular signaling traces the recreational riverside trajectory, completing **100% closed-loop tracking of full-day green transit**!

8.3 Hard Ground Truth Anchoring Calibration

Historically, cellular signaling analytics suffered from a notable vulnerability: estimation accuracy hovered around ~70%, plagued by carrier market share expansion biases and roaming user noise.
Our system introduces a groundbreaking engineering paradigm: **【Inversely Calibrate Cellular Signaling Using Hard Ground Truth Anchors】**.



Three Core Inverse Calibration Mechanisms:

1. **Dynamic Expansion Factor Tuning**:
   Carriers conventionally apply a fixed uniform multiplier (e.g., 2.63× for a 38% market share).

However, at specific stations (e.g., Zhongli Station with high migrant worker densities, or Zhubei Station with high corporate executives), carrier market shares deviate wildly. By using transit turnstile census counts as the ground truth denominator, the system **dynamically recalculates the actual cellular sampling ratio for that spatial-temporal grid**, eliminating expansion distortion.

2. **Map-Matching Ground Truth Snapping**:
   When cellular signaling hesitates between elevated National Freeway 1 and the underground Zhonghe-Xinlu Metro line, a simultaneous card swipe at Dingxi Station causes the system to **automatically snap 100% of the signaling weight to the subterranean rail network**, instantly dispelling spatial route hallucinations.
3. **Privacy-Preserving Spatiotemporal Co-occurrence**:
   Zero personal names, national ID numbers, or phone numbers are ever exchanged. Operating on feature layers within a Federated Learning architecture, the system aligns only temporal minute intervals and spatial gantry/turnstile coordinates, achieving behavioral synthesis strictly behind privacy compliance boundaries.

8.4 Policy & Commercial Decision-Making Selection Matrix

In practice, public agencies and enterprise buyers frequently struggle with data procurement choices: "Which data should we purchase for our specific needs?" This handbook provides an authoritative decision guide:

Analytical Objective	Optimal Data Source Combination	Engineering Rationale
TPASS Budget Subsidy Audit	**TICP Ticketing Data (100% Mandatory)**	Legal financial audit contracts and statutory fare types are mandatory for Treasury write-offs.
Freeway Bottleneck & EV Siting	**eTag Gantry Flows + TDX Parking APIs**	0m chainage precision and statutory 5-class vehicle breakdown are required for capacity engineering.
National Scenic Park Tourism	**Cellular Signaling + TDX Scenic CCTV**	Captures ticketless open-air tourists, scenic dwell times, and visual ground truth.
TOD Commercial & Retail Planning	**Ticketing (Anchors) + Cellular (Catchments)**	Anchors station exit counts while mapping surrounding street shopping and walking radii.
Islandwide Multi-Modal Split	**TICP (Rail/Transit) + eTag (Highway)**	Purest mutually exclusive combination; streams add together directly with zero double-counting.

Figure 8.1: Live Application Interface for Act III Data Strategy & Selection Guide. Clearly categorizing four operational scenarios: "Prioritize Transit Ticketing (TICP)", "Prioritize Freeway eTag Flows", "Prioritize Cellular Signaling", and "Strongly Recommend Three-Source Fusion (TOD Door-to-Door Closed Loop)".

8.5 Hands-on Operational Walkthrough

OFFICIAL LIVE WEB APPLICATION PORTAL (CLICK TO OPEN)

Multi-Source Fusion & Decision Guide →

Direct URL: https://tobytoy.github.io/taiwan-mobility-pulse/#telecom

Operational Steps:

1. Click the link above or click **【Telecom vs. Ticketing】** on the top navigation bar.
2. Navigate to **【Act II: Multi-Source Data Fusion & Door-to-Door Closed Loop】**.
3. Inspect the **5-Stage Interactive Journey Diagram** (Yonghe → Neihu Xihu).

4. Review the **"Government & Enterprise Procurement Decision Matrix"** to assess data trade-offs for your organization's projects.

8.6 Core Strategic Takeaways

> **Executive Takeaway**:
> In public policy, there is no single "magic bullet" dataset. The mark of mature data engineering is knowing which dataset provides the rigid baseline truth, which provides spatial coverage, and how to use the former to anchor and calibrate the latter.

Now that we have established the theoretical framework for national multimodal data fusion, we culminate in Chapter 9 by examining its ultimate operational manifestation: real-time scenic tourism intelligence, 4K live CCTVs, and holiday traffic management!

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Proceed to Next Chapter: Chapter 9: Scenic Tourism, Live CCTVs & Holiday Traffic Dispersal

Chapter 9: Scenic Tourism, Live CCTVs & Holiday Traffic Dispersal — Mountain-Coast Pulses & Real-Time Travel Intelligence

Language Switcher / 語言切換: [English] | 繁體中文版

> "On workdays, national highways serve as the steel arteries of industrial supply chains and commuter tides; on holidays, they become the emotional conduits of twenty-three million souls escaping toward mountains and seas. When high-speed rail terminates and off-ramp traffic overflows, how do we harness live visual streams and climatological data to master the nation's tourism pulse?"

9.1 The Nature of Tourism Mobility: From "Rigid Convergence" to "Holiday Dispersion"

In preceding chapters, we explored the "rigid mobility skeleton" shaped by daily urban commuting (twin peaks, inward radial convergence) and weekday freeway logistics. However, Taiwan's compact geography—where towering alpine peaks rise minutes from the ocean—fosters a uniquely volatile **"Tourism Spatiotemporal Tide"**:

1. **Explosive Traffic Flow Multipliers (+280% to +420%)**:
   Compared to quiet weekday baselines, traffic along National Freeway 5 (Hsuehshan Tunnel), Freeway 6 (Puli / Sun Moon Lake), Freeway 3 Nanzhou (toward Kenting), and Provincial Highway 18 (Alishan) surges to **3.2× normal volumes**, while air and ferry routes to offshore Penghu explode to **4.2×**.
2. **Full-Day Plateau Congestion & Unidirectional Tides**:
   Unlike brief commuting rush hours (07:30–09:00, 17:30–19:00), tourism exhibits a **prolonged morning outbound plateau (08:30–13:00)** and a **drawn-out afternoon return congestion (14:30–21:00)**, tightly constrained by high-occupancy vehicle (HOV) restrictions, ramp metering, and single-lane mountain passes.
3. **Extreme Weather Sensitivity**:
   Leisure travel depends on weather to an extent far exceeding work commuting. A single afternoon convective storm or coastal monsoon swell instantly reroutes Wai'ao surfers, Hehuanshan visitors, and Penghu island-hoppers.

To liberate planners and travelers from relying on delayed surveys or guesswork, **v1.3.1** deploys a four-dimensional national fusion framework:

【TDX Tourism APIs × CWA CODiS 30-Year Climatology × Tourism Administration 74 4K/360° Live CCTVs × Freeway Bureau eTag Holiday Traffic Data】.

9.2 Cross-Source Fusion Architecture: Scenic CCTV, Climatology, and Arteries

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    classDef etag fill:#059669,stroke:#10b981,stroke-width:2px,color:#fff;
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        A2["Attraction Spatial Coordinates & POIs"]:::tdx
        A3["92 TaiwanTrip Tourist Bus Routes"]:::tdx
    end

    subgraph S2["CWA CODiS Climatological Big Data"]
        B1["30-Year Observed Station Temperatures"]:::codis
        B2["Precipitation Probabilities & Sunshine Hours"]:::codis
        B3["Flood and Landslide Climate Alerts"]:::codis
    end

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        C3["Vehicle Mix (Tour Coaches 41 / Private Autos 31)"]:::etag
    end

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    B1 & B2 & B3 --> D
    C1 & C2 & C3 --> D

    D --> E1["🗺️ GIS Interactive CCTV Layer & Instant Preview"]
    D --> E2["📊 Climate Comfort vs. Highway Congestion Matrix"]
    D --> E3[" Immersive 4K/360° Real-Time Audio/Video Stream"]
```

Data Alignment Specifications

Dimension	Data Source	Scale	Key Attributes & Metrics
Live Sight	MOTC Tourism TDX Live Feed Service	74 Scenic CCTVs Islandwide	`Location`, `LiveURL`, `IFrameURL`, `ViewType` (360° Panoramic / PTZ / Fixed), `Status`
Climate Pulse	CWA CODiS Weather Database	Nearest 30-Year Weather Station	Seasonal temperatures, summer cooling comfort, annual rain days, best travel months, storm warnings
Mobility Arteries	Freeway eTag + TaiwanTrip	114 Dynamic Corridors, 92 Bus Routes	Key highway corridors (Hwy 18, Fwy 5, Hwy 11), holiday surge multipliers, peak hours, THSR/TRA links

Figure 9-1: Live Application Interface — Tourism Destinations & Live CCTV Dashboard spanning six geographical clusters, CODiS historical travel windows, eTag holiday surge multipliers, and TaiwanTrip bus connections.

9.3 In-Depth Mobility Profiles across Six Scenic Clusters

Figure 9-2: Live Application Interface — Immersive 4K / 360° Live Streaming Window (Matsu Nangan Iron Fort demo), featuring an ad-free stream paired with real-time weather and highway transit sidebars.

1. Chiayi Alishan Alpine Sea of Clouds Corridor (11 Live Feeds)

- **Flagship Spots**: Eryanping Trail, Taiping Suspension Bridge, Fenqihu Old Street, Sheng-Le Farm, Lijia Station, Niupuzi Prairie.
- **CODiS Climate**: Alishan Station (elevation 2,213m), annual mean temperature 11.8°C (summer mean 14.5°C), making it Taiwan's premier summer retreat; autumn and winter feature dry, crisp air with optimal visibility for cloud seas.
- **eTag Highway Artery**: Freeway 3 Zhongpu Interchange → Provincial Highway 18 (Alishan Hwy). Weekend traffic surges by **+340%**, outbound peak between 08:30–11:30; afternoon mountain fog triggers early downhill return queues from 14:00.
- **Transit Recommendation**: THSR/TRA Chiayi Station connecting to "TaiwanTrip Alishan Route (7322 / 7329)", offering dedicated right-of-way without mountain driving fatigue.

2. Nantou Sun Moon Lake Multimodal Cluster (7 Live Feeds)

- **Flagship Spots**: Ita Thao Pier, Wenwu Temple, Nine Frogs Stack, Jinlong Mountain Sunrise, Ci'en Pagoda, Dakeleha Overwater Bikeway.
- **CODiS Climate**: Sun Moon Lake Station (elevation 1,015m), annual mean temperature 19.5°C; autumn (Sept–Nov) is the most pleasant season for the International Fireworks & Music Festival. The Nine Frogs Stack serves as an empirical indicator of dry-season water storage.

- **eTag Highway Artery**: Freeway 3 Wufeng Junction → Freeway 6 Ailan Interchange → Highway 21. Holiday traffic swells to **3.1× normal volume**, with off-ramp backups at Ailan regularly extending onto the freeway before noon.
- **Transit Recommendation**: Take "TaiwanTrip Sun Moon Lake Route (6670)" from THSR Taichung Station directly to Shuishe Pier and lakeside loop bus (6669).

3. Northeast Coast, North Coast & Yilan Surfing Corridor (15 Live Feeds)

- **Flagship Spots**: Wai'ao Beach (facing Guishan Island), Yehliu Geopark, Fulong Beach, Zhongjiao Bay, Laomei Green Reef, Bitou Cape.
- **CODiS Climate**: Fugueijiao and Suao Stations, annual mean temperature 22.1°C. Winter brings the northeast monsoon, while summer transforms the coast into a water sports haven; Laomei Green Reef is a seasonal phenomenon limited to March–May.
- **eTag Highway Artery**: Freeway 5 Hsuehshan Tunnel → Toucheng Interchange → Highway 2 Coastal Road. Saturday morning southbound delays and Sunday afternoon northbound queues extend up to 8 km, with traffic volume exploding **+380%**.
- **Transit Recommendation**: Ride TRA Yilan Line Fast Locals directly to Toucheng/Wai'ao, or connect via TaiwanTrip Coastal Dunes Route (Green 18) and Crown North Coast Route (716).

4. East Rift Valley, Coastal Artery & Green Island (14 Live Feeds)

- **Flagship Spots**: Sanxiantai Arch Bridge, Jialulan Recreation Area, Jinzun Surf Spot, Luye Highland (Hot Air Balloon Festival), Liushidan Mountain Daylilies, Shitiping, Green Island Fanchuanbi.
- **CODiS Climate**: Chenggong and Taitung Stations, annual mean temperature 23.8°C; a warm maritime climate with virtually no winter. Summer and autumn require vigilance for Pacific typhoon landfalls.
- **eTag Highway Artery**: Highway 9 (Valley) and Highway 11 (Coast). Holiday cross-county road trips surge by **+260%**.
- **Transit Recommendation**: TRA Eastern Mainline (EMU3000 New Ziqiang) to Taitung Station, transferring to "TaiwanTrip East Coast Route (8101A)".

5. Southern Sunshine, Kenting & Dapeng Bay (12 Live Feeds)

- **Flagship Spots**: Kenting Nanwan, Baisha Bay, Eluanbi Lighthouse, Jialeshui, Dapeng Bay Sailing Base, Liuqiu Island Beauty Cave.
- **CODiS Climate**: Hengchun Station, annual sunshine hours exceed 2,048 hours; winter features strong downhill katabatic winds (Luoshanfeng), while summer delivers quintessential tropical sunshine.
- **eTag Highway Artery**: Freeway 3 Nanzhou Interchange → Highway 1 / Highway 26. Surges by **+310%**, with off-ramp congestion peaking Friday evening and Sunday afternoon.
- **Transit Recommendation**: Board "TaiwanTrip Kenting Express (9189)" from THSR Zuoying Station, reaching Kenting in 120 minutes via highway express lanes.

6. Offshore Archipelagos: Matsu, Kinmen & Penghu (15 Live Feeds)

- **Flagship Spots**: Matsu Nangan Iron Fort, Beigan Qinbi Village, Dongyin Lighthouse, Kinmen Gugang Lake, Penghu Guanyinting (Fireworks Festival), Erkan Historic Village.

- **CODiS Climate**: Island maritime weather; March–May brings spring advection fog disrupting flight schedules, while summer offers vibrant maritime tourism.
- **eTag & Mobility**: Air and sea passenger volumes swell by **+420%** during peak summer festival dates.
- **Transit Recommendation**: Island public buses paired with TaiwanTrip Kinmen/Penghu routes, avoiding rental scooter exhaustion under extreme sun.

9.4 Hands-on Operational Walkthrough: CCTV & Tourism Hub

OFFICIAL LIVE WEB APPLICATION PORTAL (CLICK TO OPEN)

Scenic Destinations & Live CCTV Dashboard →

Direct URL: https://tobytoy.github.io/taiwan-mobility-pulse/#tourism

Operational Steps:

1. Click the link above or click 【**Scenic CCTV (74)**】 on the top navigation bar.
2. **Filter by Region & Visual Mode**:
   Click the cluster buttons (e.g., **"Chiayi Alishan"**, **"Yilan & Northeast Coast"**, or **"Offshore Islands"**) to filter the 74 destinations.
   Toggle between **"360° Panoramic"**, **"Pan-Tilt-Zoom (PTZ)"**, and **"Fixed Lens"**.
3. **Open the Immersive Live Stream**:
   Click on the **" Watch Live Stream"** button on any card.
   The modal window streams an ad-free 4K YouTube Live feed, accompanied by local CODiS weather metrics, eTag holiday surge multipliers, and TaiwanTrip bus connections.
4. **Switch to GIS Map CCTV View**:
   Navigate to 【**GIS Map**】 → 【**Dynamic FlowMap**】 and toggle the **" 74 Scenic CCTVs"** button on the control bar. Golden camera pins will appear across Taiwan, allowing you to preview scenic feeds directly on the map alongside dynamic traffic flows!

Figure 9-3: Live Application Interface — FlowMap featuring the golden CCTV layer activated, displaying 74 live camera pins across Taiwan alongside dynamic transit particles.

9.5 Public Policy & Smart Travel Insights

> **Smart Mobility Takeaway**:
> Traditional holiday traffic management relied on static roadside variable message signs (CMS). By fusing live 4K scenic video feeds with highway congestion surge factors and real-time public transit routes, travelers can verify destination weather and crowd densities before embarking, shifting discretionary trips toward green transit or off-peak hours and achieving distributed traffic equilibrium.

Monograph Conclusion: The Living Pulse of Taiwan

Over 9 macro chapters, we have journeyed from the foundational data philosophy of rigid ticketing baselines and zero-drift highway sensors, through 114 dynamic corridors, off-ramp catchment halos, micro-personas, 30-year climate trends, and TOD commercial hubs, culminating in live 4K scenic video streams.

Taiwan's mobility pulse is neither a static map nor cold numbers in an audit report. It is the living, synchronized rhythm of 23 million people interacting with their island's mountains, coasts, cities, and

weather. May this interactive monograph serve as an illuminating benchmark for intelligent transportation engineering, urban spatial science, and national data governance!

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